

Supported Model Replacement Across Target Prevalences

A Paired Replication-Based Method for Average Precision

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Methodological proposal

Abstract

A decision about whether to replace an incumbent model with a candidate often begins with an observed difference in a chosen performance measure. When the candidate’s advantage is intended to support replacement across target conditions or empirical evaluations, a further question arises: how much of that advantage remains when the replacement claim is subjected to those declared challenges? We address this question with a falsification-oriented support construction defined on paired, condition-indexed evaluation effects.

The construction yields two complementary summaries: supported magnitude, which describes the effect retained under the challenge, and literal survival of the claimed advantage, which records its persistence above a declared floor. The condition set, evaluation regime, and challenge order determine the scope and severity of the assessment.

We develop the construction for model replacement using a prior-standardized and chance-normalized form of average precision. The framework separates evidence supplied by distinct evaluations from computational projections based on resampling a single evaluation. The proposed finite-evidence replacement rule requires supported magnitude and literal survival to exceed prespecified thresholds. The construction can be applied to other performance measures, such as AUROC, once their relevant effects and conditions of challenge have been specified.

Complementary empirical illustrations apply the projected path to one temporal Bank Marketing evaluation and the observed path to 91 ProteinGym assays. The latter comparison finds no stable replacement ordering among EVE, ESM-1v, and ESM-2 under the declared assessment.

1 Performance Claims and their Support

Prediction systems often rank cases so that a selected set can receive further attention. A radiology service, for example, may send its highest-scored images for review. The composition of that set matters, not only the separation between outcome classes, because the selected set is what the decision maker receives.

Separation (eg AUROC) and selected-set purity (eg Average Precision) part company when prevalence changes. Suppose a threshold selects 80% of positive cases and 5% of negative cases. In a balanced population of 10,000 cases, it returns 4,000 positives and 250 negatives, for precision 0.94. In a population with 1% positives, the same threshold returns 80 positives and 495 negatives, for precision 0.14. Its true-positive and false-positive rates have not moved. The supply of negative cases has, and that is enough to change the meaning of the selected set.

Average precision aggregates precision as the ranking is traversed, so ordinary empirical AP evaluates performance using the positive fraction in the evaluation data. Let D denote those data:

the observed outcomes and model scores on a common set of N evaluation units. Thus N is the number of data points on which AP is calculated. When D represents one intended population, its prevalence can remain implicit. Reporting AP at the evaluation’s own prevalence answers a question about that sample. Resampling to a target ratio discards data while still fixing a single population. Reporting the precision–recall curve defers the choice to a reader who lacks the sampling information needed to make it. Each leaves the claim indexed to one population. A first task may therefore be to construct a target-indexed quantity, $AP_{\pi}(D)$, that allows for calculation of average precision at any chosen prevalence using all of the available data. Such a quantity carries the selection rates observed in the evaluation over to the target population unchanged, which is a claim about that population rather than a property of the arithmetic.

One target-prevalence value does not render a performance claim applicable to several different deployment populations. Any single convenient prevalence hides where performance weakens, while averaging over a distribution of target prevalences both requires that distribution to be defensible and permits a strong result at one prevalence to offset a weak result elsewhere. The claim pursued here is the more demanding one: that a performance level holds throughout a scientifically plausible prevalence range rather than in expectation across it.

A second problem arises if the claim is intended to persist under new evaluation units or under other sources of variation. This calls for an evaluation regime that specifies which empirical tests fall within the scope of the claim. The extent to which a claim survives those attempts at refutation is what will be called its support.

The population and replication problems cannot be solved independently. Different evaluations may weaken at different prevalences, so challenging every evaluation at one common prevalence can miss the condition that defeats each of them. A claim required to hold throughout the population set in every evaluation must allow the limiting prevalence to change between evaluations.

Neither problem concerns the precision of an estimate. Variation under a declared computational resampling law describes sensitivity to recombinations of the observed units; it does not say whether the advantage would remain in a population with a different supply of negatives, or in the next evaluation. What is at issue is survival rather than precision: whether the advantage remains once the conditions under which it could fail have been named and imposed.

Call D a full evaluation when it contains the outcomes and scores needed to evaluate every declared condition on one set of units, so that no condition requires a separate dataset. Let

$$\mathcal{A} = (\mathcal{C}, \mathcal{R}, K),$$

where $\mathcal{C}$ contains the conditions that can be evaluated within each full evaluation, $\mathcal{R}$ specifies the admissible full evaluations, and K is the order of the replication challenge. For AP, target prevalence will supply the condition set. Changes whose effects cannot be evaluated by varying prevalence must be represented by distinct evaluations admitted by $\mathcal{R}$. The order K is the number of distinct evaluation effects in each subset used by the replication challenge. It is neither estimated from data nor equipped with a default. It states how severe a test the claimant has chosen to face, while the number of observed evaluations states how much empirical evidence is available. An observed order- K assessment therefore requires at least K separately conducted evaluations. If fewer are available, that assessment is unavailable; K should not be lowered after the results are seen merely to obtain a verdict. These elements belong to the claim rather than to the analysis. Like a preregistered analysis plan, they lose their force if they are chosen after the comparison has been seen.

Probability measures introduced below are counterfactual or computational characterizations. They are not asserted to be data-generating laws for the observed units or for unseen evaluations. The support construction instead acts directly on a finite indexed effect list. A target distribution states a declared population description, and a resampling law defines a computational projection.

What can then be assessed depends on the evidence available. Resampling a single evaluation supplies a computational projection under a declared resampling procedure. It can recombine only the case types, scores, and failure mechanisms represented in that evaluation. The resulting support is projected rather than observed, and no number of resamples converts one evaluation into new empirical tests.

The motivating application we develop is model replacement: whether a candidate model A should replace an incumbent model B on ranking performance. Pairing introduces a third difficulty. Subtracting two separately challenged reports lets each model meet its adverse condition in a different population or a different evaluation. Forming the difference on the same units removes that latitude at the cost of a new one, since finite-sample AP can reward whichever model divides tied scores more finely, whether or not the division tracks the outcomes.

Section 6 later contrasts a one-evaluation computational projection with an observed finite list of DMS assay effects. The examples are used to expose the operator’s mechanics and evidential scope, not to infer a distribution of unseen evaluations.

2 Average precision across target prevalences

2.1 Prior-standardized average precision

Changing the observed positive fraction can alter both the population represented and the evidence used to describe the ranking. Prior standardization separates these operations by asking a counterfactual question: what precision would the observed class-conditional ranking behavior produce under a different class prior?

Within one evaluation, let S_j be the score produced by the model instance used for $j \in \{A, B\}$. That instance and its scoring rule remain fixed throughout prior standardization. Writing S generically for S_A or S_B , let the observed class-conditional selection rates at threshold τ be

$$r(\tau) = \mathbb{P}_{\text{obs}}(S \geq \tau \mid Y = 1), \quad f(\tau) = \mathbb{P}_{\text{obs}}(S \geq \tau \mid Y = 0).$$

Here $\mathbb{P}_{\text{obs}}$ characterizes the observed score–outcome distribution, while $\mathbb{P}_{\text{target}}$ below characterizes a declared target population. Neither notation posits a stochastic law that generated the evaluation data. Holding these rates fixed while varying the target prevalence is warranted only when the class-conditional score behavior transports. Because the two models are compared on the same units, the transport condition for each target population must preserve their joint class-conditional score behavior:

$$\mathbb{P}_{\text{obs}}\{(S_A, S_B) \in G \mid Y = y\} = \mathbb{P}_{\text{target}}\{(S_A, S_B) \in G \mid Y = y\}, \quad y \in \{0, 1\}, \quad (1)$$

for every measurable $G \subseteq \mathbb{R}^2$. This condition entails the corresponding marginal invariance for S_A and S_B , which supplies each model’s target AP profile. It also preserves the within-class association between the scores in paired computational projections. This score-level form of prior-probability shift is weaker than invariance of $P(X \mid Y)$ but follows from it for fixed scoring rules [17, 20, 10].

Under this condition, Bayes' rule expresses precision at threshold τ in a target population with prevalence $\pi \in (0, 1)$ as

$$p_\pi(\tau) = \frac{\pi r(\tau)}{\pi r(\tau) + (1 - \pi)f(\tau)}. \quad (2)$$

Its odds separate the target prior from the ranking behavior:

$$\frac{p_\pi(\tau)}{1 - p_\pi(\tau)} = \frac{\pi}{1 - \pi} \frac{r(\tau)}{f(\tau)}. \quad (3)$$

The functions $r(\tau)$ and $f(\tau)$ are therefore held at the values supplied by evaluation D while π indexes the declared target populations. Varying π changes only the class prior. Aggregating $p_\pi(\tau)$ across the distinct score thresholds using the observed positive-class recall increments gives $\text{AP}_\pi(D)$, prior-standardized AP for evaluation D at prevalence π [7, 18]. Appendix A gives the empirical definition.

The class-conditional score-transport condition matters because prevalence can accompany other population changes. A referral population may contain more severe cases within each outcome class, while a new measurement process may change the score for otherwise similar cases. In either setting, Equation (2) reweights rates that need not describe the target. Those changes must be represented by observed evaluations from the corresponding environments rather than by prior adjustment alone [2].

2.2 A common effect scale

Prior-standardized AP still has a different random-ranking value at every target prevalence. Population random ranking has $r(\tau) = f(\tau)$ and hence $\text{AP}_\pi = \pi$. Define

$$\text{CNAP}_\pi(D) = \frac{\text{AP}_\pi(D) - \pi}{1 - \pi}. \quad (4)$$

CNAP is zero for population random ranking, one for perfect ranking, and negative below chance. The normalization has the chance-corrected form (observed – chance)/(ceiling – chance) [6]. For two models,

$$\text{CNAP}_{A,\pi} - \text{CNAP}_{B,\pi} = \frac{\text{AP}_{A,\pi} - \text{AP}_{B,\pi}}{1 - \pi}. \quad (5)$$

The common chance term cancels, while division by the random-to-perfect headroom $1 - \pi$ expresses the AP gap on a common opportunity scale. Thus a margin d requires a raw AP gain of $d(1 - \pi)$ at prevalence π ; the same identity applies after tie averaging. This scaling is substantive and is appropriate when the replacement margin refers to a fraction of random-to-perfect improvement rather than an equal raw AP increment at every prevalence.

Common endpoints do not make the profile prevalence-invariant. At a threshold with true-positive and false-positive rates r_h and f_h , the normalized contribution is

$$\frac{p_{\pi,h} - \pi}{1 - \pi} = \frac{\pi(r_h - f_h)}{\pi r_h + (1 - \pi)f_h}. \quad (6)$$

Rare-positive populations therefore charge false positives more heavily. Two models can exchange order as π changes even when both profiles improve with prevalence.

Let $\Pi_\star \subset (0, 1)$ be a nonempty compact set chosen from deployment plans or scientific knowledge before the paired-difference profile is examined. The CNAP profile then supplies the AP-specific

condition index: $c = \pi$ and $\mathcal{C} = \Pi_\star$. No prevalence has yet been selected or averaged away. The next problem is to determine what a claim throughout this profile must retain when the evaluation is replicated.

3 Support across declared conditions and evaluations

The prevalence profile in Section 2 is one instance of a condition-indexed effect. More generally, let $\Delta_c(D)$ denote an effect under condition $c \in \mathcal{C}$, oriented so that larger values favor the claim. Every condition in the nonempty set $\mathcal{C}$ must be evaluable from one full evaluation D . For AP under score-level prior invariance, c is a target prevalence and $\mathcal{C} = \Pi_\star$. Section 4 constructs the paired, tie-averaged CNAP difference that supplies $\Delta_\pi(D)$ for model replacement.

Two sources of challenge enter through different empirical objects. Conditions in $\mathcal{C}$ vary within each full evaluation. Changes that alter class-conditional ranking behavior require full evaluations conducted in the corresponding contexts. The evaluation regime states which of those evaluations address the declared claim.

3.1 The evaluation regime

Let $\mathcal{R}$ specify the requirements under which a full evaluation D counts as an admissible empirical test of the claim. These requirements include the object under test, eligible populations, measurement standards, available outcomes and scores, and the independent evaluation unit. The regime defines scientific scope. It makes no assertion that admissible evaluations share a statistical distribution. Statistical descriptions applied after observation may differ between contexts.

A separately conducted full evaluation satisfying $\mathcal{R}$ is an *empirical replication*. Replications use nonoverlapping evaluation units. Their model construction and data collection are not altered in response to another evaluation in the same replication set. Observations within an evaluation may be clustered or matched when the design and analysis preserve that structure. Administrative partitions, resampling draws, and cross-validation folds do not create empirical replications.

A fixed-model regime holds the fitted models and measurement process fixed while admitting new evaluation units. A broader regime may admit distinct environments or a prespecified model-updating procedure. When models are updated, the common object under test is the procedure that produces them; information used for updating must remain separate from the subsequent evaluation. In that setting, A and B denote procedures whose context-specific instances supply the scores in each evaluation. If an earlier evaluation prompts a later update, the evaluations form an adaptive evidential sequence. They do not constitute a set of separate empirical replications under the definition above, and their order and information flow require explicit analysis.

Several empirical replications may occur in different domain contexts while remaining admissible tests of the same scoped claim. Their identity as separate tests should be retained when contextual variation is scientifically meaningful. If dataset membership is only administrative and all units address one aggregate population, pooling may be more appropriate. Scientific reach is determined by $\mathcal{R}$ and by the empirical tests conducted within it.

3.2 The effect retained across all conditions

Let $D_1, \dots, D_L$ be a nonempty finite list of full evaluations admitted by $\mathcal{R}$. A claim throughout $\mathcal{C}$ requires one effect level to hold under every declared condition within each evaluation. For evaluation D_ℓ , the greatest such level is

$$z_\ell = \inf_{c \in \mathcal{C}} \Delta_c(D_\ell). \quad (7)$$

This is the largest uniform lower bound on the condition profile: every declared condition retains at least z_ℓ , and no larger level has that property. The infimum covers finite condition sets and continuous profiles whose lowest value may fail to be attained. An average across conditions would require weights over $\mathcal{C}$ and would permit favorable conditions to compensate for unfavorable ones.

This operation is algebraically related to an intersection bound. Its role here is different: Equation (7) is evaluated exactly on the condition profile supplied by one finite evaluation, rather than used to estimate an infimum of population-indexed functions. Section 7.2 develops this distinction.

Suppose three evaluations have the profiles

	$\pi = 0.01$	$\pi = 0.40$	z_ℓ
D_1	-0.10	0.30	-0.10
D_2	0.20	-0.20	-0.20
D_3	0.30	0.30	0.30

At the two common prevalences, the mean effects are $0.133\bar{3}$ and $0.133\bar{3}$. Equation (7) gives $z_1 = -0.10$, $z_2 = -0.20$, and $z_3 = 0.30$ because the limiting prevalence differs between the first two evaluations.

Collect the retained effects in $\mathbf{z} = (z_1, \dots, z_L)$. The replication challenge determines how strongly low values in this finite list influence the assessment.

3.3 Support across retained evaluation effects

The empirical mean of $\mathbf{z}$ permits a large retained effect in one evaluation to offset a small or negative retained effect in another. Its minimum gives the level retained by every available evaluation. The order K supplies graded lower-tail emphasis between these endpoints.

Definition 1 (Replication-survival support). *For $1 \leq K \leq L$, let*

$$\mathcal{I}_{L,K} = \{I \subseteq \{1, \dots, L\} : |I| = K\}.$$

Define

$$S_{K,\mathcal{C}}(\Delta; \mathbf{D}) = S_K(\mathbf{z}) = \binom{L}{K}^{-1} \sum_{I \in \mathcal{I}_{L,K}} \min_{\ell \in I} z_\ell. \quad (8)$$

$S_K(\mathbf{z})$ is the average weakest effect over all subsets of K distinct evaluation indices. Each subset receives equal weight, and no evaluation is repeated within a challenge. Equal numerical effects can still represent distinct empirical tests. This construction supplies no probability law for unseen

evaluations. A different weighting would define a different assessment and would require a scientific basis.

At $K = 1$, support is the empirical mean of the retained effects. At $K = 2$,

$$S_2(\mathbf{z}) = \bar{z} - \frac{1}{L(L-1)} \sum_{1 \leq i < \ell \leq L} |z_i - z_\ell|. \quad (9)$$

The second-order criterion is the mean retained effect minus half the mean pairwise disagreement over distinct ordered pairs. As K increases, lower retained effects receive more weight. The endpoints are $S_1(\mathbf{z}) = \bar{z}$ and $S_L(\mathbf{z}) = \min_\ell z_\ell$.

For the retained-effect list $(0.70, 0.35, 0.05)$, $S_1 = 0.366\bar{6}$. The order-2 construction contains three equally weighted subsets:

$$\{0.70, 0.35\}, \quad \{0.70, 0.05\}, \quad \{0.35, 0.05\}.$$

Their minima are 0.35, 0.05, and 0.05, giving

$$S_2 = \frac{0.35 + 0.05 + 0.05}{3} = 0.15.$$

The sole order-3 subset contains all three effects, so $S_3 = 0.05$. Thus the sequence $0.366\bar{6}, 0.15, 0.05$ moves from the mean to the observed minimum. The list length records the three available effects; K controls how many distinct effects must jointly face each challenge.

For any nonempty finite effect list $\mathbf{t} = (t_1, \dots, t_L)$ and $1 \leq K \leq L$, $S_K(\mathbf{t})$ denotes the same distinct-subset construction with $\mathbf{t}$ in place of $\mathbf{z}$.

3.4 Literal survival at a declared floor

Supported magnitude retains the numerical size of every subset minimum. A decision may also require the retained effect to clear a prespecified floor d . Let

$$R_d = \sum_{\ell=1}^L \mathbf{1}\{z_\ell > d\}.$$

Empirical literal survival is

$$\psi_{K,d}(\mathbf{z}) = \frac{\binom{R_d}{K}}{\binom{L}{K}}, \quad (10)$$

where $\binom{R_d}{K} = 0$ when $R_d < K$. It is the fraction of distinct order- K subsets whose effects all clear d .

For any level s , define

$$R_s = \sum_{\ell=1}^L \mathbf{1}\{z_\ell > s\}, \quad z_- = \min_\ell z_\ell, \quad z_+ = \max_\ell z_\ell.$$

Then

$$\begin{aligned} S_K(\mathbf{z}) &= z_- + \int_{z_-}^{z_+} \psi_{K,s}(\mathbf{z}) ds \\ &= z_- + \int_{z_-}^{z_+} \frac{\binom{R_s}{K}}{\binom{L}{K}} ds. \end{aligned} \quad (11)$$

At each level s , the integrand is the fraction of distinct order- K subsets whose effects all clear s . Supported magnitude integrates this finite-list survival fraction over all effect levels. Literal survival evaluates it at d . For a generic finite list $\mathbf{t}$, the notation $\psi_{K,d}(\mathbf{t})$ applies the same construction. Appendix B.2 gives the ordered representation and further details.

3.5 Observed and projected effect lists

Let N count the evaluation units whose outcomes and scores enter an AP calculation, M count separately conducted empirical evaluations, and J count computational replications.

The generic list length L in the preceding definitions can then be instantiated as either M or J depending on the nature of the evaluations. For example, for projected effects from J computational replications of one observed evaluation, $L = J$ and the list index identifies the replications. In either case, the order must satisfy $K \leq L$.

When M separately conducted full evaluations satisfy $\mathcal{R}$, their retained effects form

$$\mathbf{z}_M^{\text{obs}} = (z_1, \dots, z_M).$$

When $K \leq M$, applying Equations (8) and (10) directly to this indexed list gives observed support and literal survival. If $M < K$, the declared observed assessment is unavailable because the evidence does not contain enough distinct evaluations to form an order- K subset.

One observed evaluation has not empirically faced several replications. Resampling it generates a list of J computational effects under a declared resampling procedure:

$$\mathbf{z}_J^{\text{proj}} = (z_1, \dots, z_J).$$

When $J \geq K$, applying the same finite-list formulas gives *projected support* and projected literal survival. These quantities characterize the effects produced by the procedure from the observed evaluation. The procedure uses only score values and failure modes represented there, so its challenge is narrower than one supplied by new empirical replications. In practice, J is usually chosen much larger than K to control Monte Carlo error.

Separate empirical replications can reveal failures arising from variation within the scope of $\mathcal{R}$ that the original evaluation did not contain. An evaluation involving a mechanism excluded from $\mathcal{R}$ may show that the declared assessment was too narrow.

Increasing J reduces Monte Carlo error in the same projected quantities. Increasing K changes the challenge applied to either effect list. Increasing M changes the empirical evidence available for observed support.

3.6 Challenge order and scope

Equation (8) first identifies the limiting condition within each evaluation and then applies the replication challenge to the retained effects. The condition attaining z_ℓ may therefore differ between evaluations.

For a fixed condition c , write $\Delta_c[\mathbf{D}] = (\Delta_c(D_1), \dots, \Delta_c(D_L))$ for the effect list obtained by applying that condition to every evaluation. Reversing the order gives

$$\inf_{c \in \mathcal{C}} S_K(\Delta_c[\mathbf{D}])$$

and requires one condition to challenge all evaluations before the limiting condition is selected. In the profiles displayed above, D_1 is weakest at $\pi = 0.01$, D_2 is weakest at $\pi = 0.40$, and D_3 has the same effect at both prevalences. At $K = 2$, the retained-effect order gives

$$S_2(-0.10, -0.20, 0.30) = -\frac{1}{6}.$$

The common-condition values are $S_2(-0.10, 0.20, 0.30) = 0$ at $\pi = 0.01$ and $S_2(0.30, -0.20, 0.30) = -1/30$ at $\pi = 0.40$, giving $-1/30$ after the infimum. The retained-effect order supplies the stronger test within the declared assessment: each evaluation faces every declared condition, and the condition exposing its weakest effect may differ between evaluations.

The retained-effect order always satisfies

$$S_K(\mathbf{z}) \leq \inf_{c \in \mathcal{C}} S_K(\Delta_c[\mathbf{D}]). \quad (12)$$

Equality holds when the same condition minimizes every relevant profile; Appendix B gives the proof.

Enlarging $\mathcal{C}$ gives each evaluation more conditions under which its effect may be limited. Increasing K gives greater influence to the lower retained effects.

Proposition 1 (Challenge monotonicity). *If $\mathcal{C}_1 \subseteq \mathcal{C}_2$ and $1 \leq K_1 \leq K_2 \leq L$, then*

$$S_{K_2, \mathcal{C}_2}(\Delta; \mathbf{D}) \leq S_{K_1, \mathcal{C}_1}(\Delta; \mathbf{D}).$$

$\mathcal{R}$ determines which evaluations and procedures are admissible. Arbitrary changes of $\mathcal{R}$ have no comparable ordering because they change the scientific scope of the assessment.

Support has a limited negative interpretation. A level is retained when it survives a challenge constructed from the declared conditions, the admitted empirical tests, and the chosen distinct-subset order. Survival establishes no truth beyond that assessment [15, 16]. The regime supplies admissibility, the conducted evaluations supply the finite indexed effect list, and K sets the challenge applied to that list.

4 Constructing the paired AP effect

The support operator requires a finite list of evaluation-level effects. For replacement, each effect compares candidate A with incumbent B within the same evaluation and target prevalence. Its AP calculation should give no expected credit to outcome-blind refinement of score ties.

4.1 Marginal summaries do not compose

A replacement decision concerns an advantage realized under the same conditions. For $\mathbf{D} = (D_1, \dots, D_L)$, define

$$\begin{aligned} a_\ell &= \inf_{\pi \in \Pi_\star} \text{CNAP}_{A, \pi}(D_\ell), & b_\ell &= \inf_{\pi \in \Pi_\star} \text{CNAP}_{B, \pi}(D_\ell), \\ t_\ell &= \inf_{\pi \in \Pi_\star} \{\text{CNAP}_{A, \pi}(D_\ell) - \text{CNAP}_{B, \pi}(D_\ell)\}. \end{aligned}$$

Let $\mathbf{a} = (a_1, \dots, a_L)$, with $\mathbf{b}$ and $\mathbf{t}$ defined similarly. In $S_K(\mathbf{a}) - S_K(\mathbf{b})$, the adverse prevalence may differ between models within an evaluation, and the two support calculations may emphasize different evaluations. The discrepancy has a fixed direction:

$$S_K(\mathbf{a}) - S_K(\mathbf{b}) \geq S_K(\mathbf{t}). \quad (13)$$

Separate reports therefore overstate the supported paired advantage by an amount they do not identify. The proof in Appendix C uses only properties of the infimum and minimum, so the same result applies after tie averaging is introduced below.

4.2 Ordinary pairing rewards arbitrary refinement

Pairing aligns the units and outcomes, but ordinary finite-sample AP still depends on the score vectors' tie structures. Stepwise AP is upward displaced under random ordering because it evaluates precision when positive cases enter the ranking [3]. A fully tied score vector has one threshold and returns the population random-ranking value, while a vector of distinct scores creates more opportunities for favorable positive entries.

Four units show why this matters for replacement. Let model A assign four distinct scores and model B assign one common score. With two positive labels, two negative labels, and $\pi = 1/2$, assign the labels uniformly over the six possibilities. Neither score vector is associated with the outcomes. Under the ordinary threshold-block convention, model A has expected AP $49/72$, while model B has AP $1/2$ under every assignment. Hence

$$\mathbb{E}[\text{CNAP}_A - \text{CNAP}_B] = \frac{13}{36}.$$

For a $K = 2$ challenge under two independently randomized label assignments, the expected minimum of the paired differences remains positive at $29/216$. Pairing has favored the model that broke ties without information.

No single additive reference correction removes this displacement at every level of model performance. The expected paired statistic under joint label permutation is large near no association and vanishes at perfect separation, where both models reach the ceiling. Appendix C gives an exact four-unit example in which subtracting the joint-permutation expectation favors the coarser model even though both models separate the classes perfectly. Tie handling must therefore be resolved before the nonlinear population and replication challenges are applied.

4.3 Tie-averaged average precision

Let $\mathcal{O}(D)$ be the strict rankings consistent with a score vector, obtained by ordering tied units in every possible way while preserving the order of the score blocks.

Definition 2 (Tie-averaged average precision).

$$\widetilde{\text{AP}}_\pi(D) = \frac{1}{|\mathcal{O}(D)|} \sum_{o \in \mathcal{O}(D)} \text{AP}_\pi(D_o), \quad (14)$$

where D_o is evaluation D resolved into strict ranking o . Define $\widetilde{\text{CNAP}}_\pi$ through Equation (4).

The convention corresponds to a declared random or otherwise outcome-blind resolution of units within each score block. It is appropriate when an operational cutoff may split a block and the model supplies no within-block information. If deployment selects whole score blocks or uses a prespecified informative secondary rule, threshold-block AP or the resulting ordered ranking may be more faithful. Under the random-resolution regime, tie averaging is equivalent to adding independent continuous jitter within blocks and averaging ordinary AP; Appendix D evaluates that average without enumeration or random jitter.

Tie averaging still rewards informative refinement. A finer score improves realized AP when it orders cases correctly within a coarse block and loses AP when it orders them incorrectly. Only refinement unrelated to the outcomes receives zero expected credit.

Proposition 2 (Neutrality to arbitrary refinement). *Suppose model A preserves the order of model B’s score blocks and, conditionally on S_B and Y , its strict ranking is uniform over the rankings consistent with S_B . Then, for every π ,*

$$\mathbb{E}[\widetilde{\text{AP}}_{A,\pi} \mid S_B, Y] = \widetilde{\text{AP}}_{B,\pi}.$$

The proposition conditions on the realized outcomes, so it applies when the coarse model is informative. It removes expected credit for arbitrary refinement without requiring exchangeable evaluation units.

4.4 The supported paired advantage

The AP-specific condition index is the target prevalence. Define the paired effect profile

$$\Delta_\pi^{A:B}(D) = \widetilde{\text{CNAP}}_{A,\pi}(D) - \widetilde{\text{CNAP}}_{B,\pi}(D),$$

and let $\mathbf{D} = (D_1, \dots, D_L)$. The retained effects from Section 3 are

$$z_\ell^{A:B} = \inf_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D_\ell), \quad \mathbf{z}^{A:B} = (z_1^{A:B}, \dots, z_L^{A:B}).$$

The model-replacement support is the corresponding instance of Equation (8):

$$\theta_{A:B;K}(\mathbf{D}) = S_{K,\Pi_\star}(\Delta^{A:B}; \mathbf{D}) = S_K(\mathbf{z}^{A:B}). \quad (15)$$

Within each evaluation, the construction subtracts tie-averaged CNAP at each prevalence and then takes the prevalence infimum. It collects the retained advantages and applies the replication challenge. A positive value of $\theta_{A:B;K}(\mathbf{D})$ means that the average weakest paired advantage over subsets of K distinct evaluation effects is positive. It describes retained magnitude; literal survival evaluates the same effect list at a declared floor.

4.5 Orientation and no verdict

The reverse effect profile is $\Delta_\pi^{B:A}(D) = -\Delta_\pi^{A:B}(D)$. Define

$$z_\ell^{B:A} = \inf_{\pi \in \Pi_\star} \Delta_\pi^{B:A}(D_\ell), \quad \mathbf{z}^{B:A} = (z_1^{B:A}, \dots, z_L^{B:A}),$$

and

$$\theta_{B:A;K}(\mathbf{D}) = S_K(\mathbf{z}^{B:A}).$$

Negation occurs before the prevalence infimum and the replication challenge, so the reversed value is generally different from the negative of the original. The two orientations satisfy

$$\theta_{A:B;K}(\mathbf{D}) + \theta_{B:A;K}(\mathbf{D}) \leq 0. \quad (16)$$

The construction cannot support positive advantages in both directions.

Both orientations may be nonpositive because the declared challenges defeat superiority in either direction. The diagnostic range from $\theta_{A:B;K}(\mathbf{D})$ to $-\theta_{B:A;K}(\mathbf{D})$ reflects variation across target prevalences and disagreement across evaluations; it is not an uncertainty interval. Appendix C gives its characterization.

4.6 Reference behavior and design scope

Refinement neutrality compares a fine ranking with a coarser ranking it refines. Reference behavior for two arbitrary uninformative rankings is evaluated by a constructed diagnostic randomization.

Definition 3 (Exchangeable-label reference law). *Conditionally on both score vectors and the positive count N_+ , every label vector containing N_+ positives has probability $\binom{N}{N_+}^{-1}$.*

Under this law, labels read along any fixed strict ranking are uniform over the same set of arrangements. Expected tie-averaged AP therefore depends on N_+ , N_- , and π , but not on the score vector or its tie structure.

Write $\mathbb{E}_0$ for expectation when the exchangeable-label randomization is applied independently within each full evaluation, conditional on the score vectors and positive counts fixed by the design. This expectation is a diagnostic reference value, or anchor, showing whether score structure alone shifts the comparison under the randomization.

Proposition 3 (Reference behavior of paired support). *Under the exchangeable-label reference law,*

$$\mathbb{E}_0[\Delta_\pi^{A:B}(D)] = 0 \quad \text{for every } \pi \in \Pi_\star,$$

and consequently

$$\mathbb{E}_0[\theta_{A:B;K}(\mathbf{D})] \leq 0, \quad \mathbb{E}_0[\theta_{B:A;K}(\mathbf{D})] \leq 0.$$

The inequality need not be equality because the prevalence infimum and replication minima are nonlinear. They can give an uninformative paired effect a negative expected supported value. The result prevents differing tie structures from manufacturing a positive supported advantage, although the challenge itself may be conservative.

Full exchangeability is stronger than independence. Matched, clustered, and heterogeneous-risk designs usually permit only a restricted group of label arrangements, under which score vectors aligned with the strata can have different reference values. Tie averaging still removes arbitrary ordering within score blocks, and Equation (15) remains descriptive, but Proposition 3 no longer follows. Appendix C gives a matched-pair counterexample.

4.7 A two-part replacement rule

For a prespecified floor $d \geq 0$, define literal survival of the paired advantage as

$$\psi_{K,d}^{A:B}(\mathbf{D}) = \psi_{K,d}(\mathbf{z}^{A:B}). \quad (17)$$

Both quantities summarize the same replication-challenged paired effect. Supported magnitude retains the numerical size of every subset minimum. Literal survival records the fraction of distinct order- K subsets whose effects all exceed d .

Let $\delta \geq 0$ be the smallest replication-challenged CNAP advantage that would justify replacement, and let $\gamma \in (0, 1)$ be the minimum acceptable literal-survival fraction. The two-part policy requires

$$\theta_{A:B;K}(\mathbf{D}) > \delta \quad \text{and} \quad \psi_{K,d}^{A:B}(\mathbf{D}) > \gamma. \quad (18)$$

The magnitude condition prevents a negligible but broadly retained advantage from automatically justifying replacement and remains sensitive to severe evaluation effects. The survival condition prevents large favorable effects from compensating for an unacceptable empirical fraction of failure at d . Separating δ from d clarifies these questions: for example, $\delta > 0$ and $d = 0$ require a worthwhile retained advantage together with a high empirical survival fraction for positive superiority. Setting $\delta = d$ is permissible when the same practical effect level is intended for both purposes.

The severity of the literal-survival requirement depends jointly on K and γ . If R_d of M empirical effects clear d , observed literal survival is

$$\psi_{K,M,d}^{A:B,\text{obs}} = \frac{\binom{R_d}{K}}{\binom{M}{K}}. \quad (19)$$

For J computational effects, projected literal survival replaces M by J . The fraction is zero whenever fewer than K effects clear d and is one only when every effect clears d . Reports should give R_d/M for observed support or R_d/J for projected support, together with the corresponding subset survival, K , and γ .

For $K = 2$, each subset is a pair of complete effect-list entries. If the list contains observed assay effects, the pair contains two assays; if it contains projected computational effects, the pair contains two computational replications. It is not a pair of data points within an evaluation. A pair survives only when both of its retained effects exceed d . Thus a requirement near $0.9^2 = 0.81$ corresponds approximately to requiring nine tenths of the individual effects to clear the floor, with the exact finite fraction given by Equation (19). This is a policy threshold on the observed finite list, not a confidence level.

Define $\psi_{K,d}^{B:A}(\mathbf{D}) = \psi_{K,d}(\mathbf{z}^{B:A})$. The reverse decision applies the same two-part rule to $\theta_{B:A;K}(\mathbf{D})$ and $\psi_{K,d}^{B:A}(\mathbf{D})$, with orientation-specific thresholds if the consequences are asymmetric. If neither orientation satisfies both requirements, the analysis returns no verdict.

Equation (18) is a finite-evidence policy rule. It applies to the observed evaluation effects or to a declared computational projection, as identified in the report. Passing the rule supports replacement relative to those evaluations and challenges; it is not a confidence statement about a latent population quantity or about unseen evaluations. Failure of either requirement returns no verdict rather than evidence that the incumbent is superior.

The rule does not infer that future deployment will reproduce the observed effects. Using its verdict to guide deployment requires an external decision premise: that the declared condition set,

evaluation regime, and transport commitments are adequate for the contemplated use, together with acceptance of the consequences of residual failure. Those commitments belong to the claimant or decision maker. The support calculation makes them explicit but neither supplies nor certifies them.

5 Evidence and computational projection

Evidence is supplied by observed full evaluations. Resampling an evaluation transforms that evidence into computational effects under a declared procedure; it does not add evidence. The same support functionals apply to observed and computational effect lists, while their interpretations differ. When M separately conducted evaluations satisfying $\mathcal{R}$ are available, their effects give observed support through Equation (8); Appendix E.1 gives the calculation.

5.1 Projected support from one observed evaluation

Most applications begin with one observed full evaluation. Its relationship to model development and to any earlier evaluations must be reported. A declared empirical-resampling procedure can project the support calculation within the empirical world represented by that evaluation.

Let $D_{\text{obs},+}$ and $D_{\text{obs},-}$ contain the positive and negative evaluation units. Computational replication j samples the declared numbers of units with replacement within each class, keeps both model scores attached to every sampled unit, and evaluates the paired effect profile throughout $\Pi_\star$. Its retained effect is

$$\begin{aligned} z_j^{A:B} &= \inf_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D^{(j)}) \\ &= \inf_{\pi \in \Pi_\star} \{\widetilde{\text{CNAP}}_{A,\pi}(D^{(j)}) - \widetilde{\text{CNAP}}_{B,\pi}(D^{(j)})\}. \end{aligned}$$

The computational effect list

$$\mathbf{z}_J^{A:B,\text{proj}} = (z_1^{A:B}, \dots, z_J^{A:B})$$

is an indexed list of finite computational effects. Its projected supported magnitude is

$$\hat{\theta}_{A:B;K,J}^{\text{proj}} = S_K(\mathbf{z}_J^{A:B,\text{proj}}).$$

The same computational replications give

$$\hat{\psi}_{K,J,d}^{A:B,\text{proj}} = \frac{\binom{R_d^{(J)}}{\binom{J}{K}}}{\binom{J}{K}}, \quad R_d^{(J)} = \sum_{j=1}^J \mathbf{1}\{z_j^{A:B} > d\}. \quad (20)$$

The empirical resampling law is a constructed computational measure on recombinations of the observed units. The exact projected quantities are the corresponding support functionals under that law. The displayed quantities use J draws to approximate them. Increasing J refines the same computational projection and reduces its Monte Carlo variation. It supplies no additional empirical opportunity to reveal failure in a new context.

Stratification fixes the positive and negative counts in the projected calculation. A projection of the observed evaluation uses its class counts. A prospective projection uses the counts planned for a later evaluation. A procedure that varies the class counts must declare how they are resampled.

Clustered or matched studies must resample the declared design blocks rather than their component observations, which may preclude simple class-stratified resampling.

The same resampled units must be used for both models and throughout the prevalence profile. Resampling models separately would destroy pairing. Resampling separately at each prevalence would splice together values from different computational evaluations, so their minimum would no longer describe one evaluation’s entire prevalence profile.

The empirical-resampling projection represents only recombinations available in the observed score pools. It cannot produce unseen score values or failures caused by a new environment, measurement process, or development run. Its result must be reported as projected fixed-model support. It records no survival of K empirical replications.

5.2 Lower-tail sensitivity and sparse outcomes

The role of K creates a computational and interpretive tension. After sorting J projected computational effects, Equation (8) assigns $z_{(i)}$ weight

$$w_{i,K,J} = \frac{\binom{J-i}{K-1}}{\binom{J}{K}}, \quad i = 1, \dots, J - K + 1,$$

with zero weight on later order statistics. For M observed empirical effects, the same formula replaces J by M . Increasing K moves weight toward the lower observed effects. The stronger challenge consequently depends more heavily on whether the empirical evaluations, or the resampling collection, contain adverse evaluation effects. This dependence is especially consequential when positive outcomes are sparse and AP is coarse. Literal survival introduces an additional boundary sensitivity: when several z_ℓ lie near d , small perturbations can change the exceedance indicator even when the effect changes little.

5.3 Prevalence search and nonsmooth minima

A finite set $\Pi_\star = \{\pi_1, \dots, \pi_G\}$ defines an exact finite challenge. A grid used to approximate an interval can miss an interior minimum and raise the reported value. A one-dimensional optimizer can instead search a compact interval, with its tolerance checked against a dense grid.

Nonunique or moving prevalence minimizers require a further distinction. Uniformly accurate profiles may still yield an accurate infimum value even when the minimizing prevalence is unstable. Because the infimum is nonsmooth at flat or multiple minima, the reported minimum and the identity of the limiting prevalence can respond differently to small numerical or data perturbations. Reports should therefore separate accuracy of the minimum value from stability of its location.

5.4 Computational uncertainty

Observed support is an exact functional of the finite observed effect list, up to numerical evaluation of the condition infimum. Projected support adds Monte Carlo approximation because only J draws from the declared empirical resampling law are used. Neither source is sampling uncertainty about a latent population parameter.

Computational diagnostics should address the numerical tolerance of the prevalence search, stability across random seeds, convergence as J increases, and sensitivity when effects lie near d . Pairing, tie averaging, and the same prevalence search must be preserved in every computational replication. Monte Carlo error may be summarized because the randomization is deliberately introduced by the algorithm; it quantifies approximation of the declared projection and supplies no claim about unseen units, contexts, or evaluations.

6 Empirical illustrations

Two empirical illustrations exercise the two evidence paths without changing the interpretation of the operator. The Bank Marketing illustration begins with one observed evaluation and constructs a projected list of computational effects. The ProteinGym illustration instead applies the operator directly to a finite list of observed assay effects. In both studies the condition set is the target-prevalence interval $\Pi_\star = [0.01, 0.50]$, the challenge order is $K = 2$, and the two-part rule uses $\delta = d = 0$ and $\gamma = 0.81$. These values define the illustrated policy; they are not proposed as defaults.

6.1 One evaluation and a computational projection: Bank Marketing

The UCI Bank Marketing data record ordered direct-marketing contacts and whether each customer subscribed to a term deposit [13]. The first 32,950 of 41,188 records were used to fit the models, and the final 8,238 formed one temporal evaluation. Contact duration was excluded before fitting because it is unavailable at the decision to initiate the contact. The primary comparison was random forest against a logistic model using the available pre-contact predictors. A full logistic model against a demographic-only logistic model supplied a positive control. The models and the split serve to illustrate the assessment rather than to make a claim about banking practice.

Figure 1 holds the fitted full-logistic ranking and its observed class-conditional score behavior fixed while varying only the declared target prevalence. AP changes with that composition. CNAP retains the profile but places population random ranking at zero throughout the interval.

The paired difference must be formed at each prevalence before the weakest condition is selected. Figure 2 shows why. The random-forest ordering changes relative to the full logistic model inside the declared interval, so its retained effect is negative even though it is positive at lower prevalences. The positive-control difference remains positive throughout the interval.

The temporal holdout supplies $M = 1$ observed full evaluation, so it cannot supply observed order-2 support. We instead drew $J = 600$ paired, class-stratified computational replications from its score pools, preserving the observed evaluation counts of 2,540 subscribers and 5,698 non-subscribers. Every replication used the same resampled units for both models and throughout the prevalence profile. Figure 3 shows the resulting projected list for the positive control. The prevalence infimum is taken within each complete replication before order-2 support is applied across replications.

For random forest against full logistic regression, projected order-2 support was -0.0104 and 114 of 600 effects cleared zero, giving literal survival 0.036 and no verdict. For full against demographic-only logistic regression, projected support was 0.0130 and all 600 effects cleared zero, giving literal survival 1.000 and a supported-replacement verdict. The reversed orientation did not satisfy the rule in either comparison. Figure 4 keeps the magnitude and survival requirements separate.

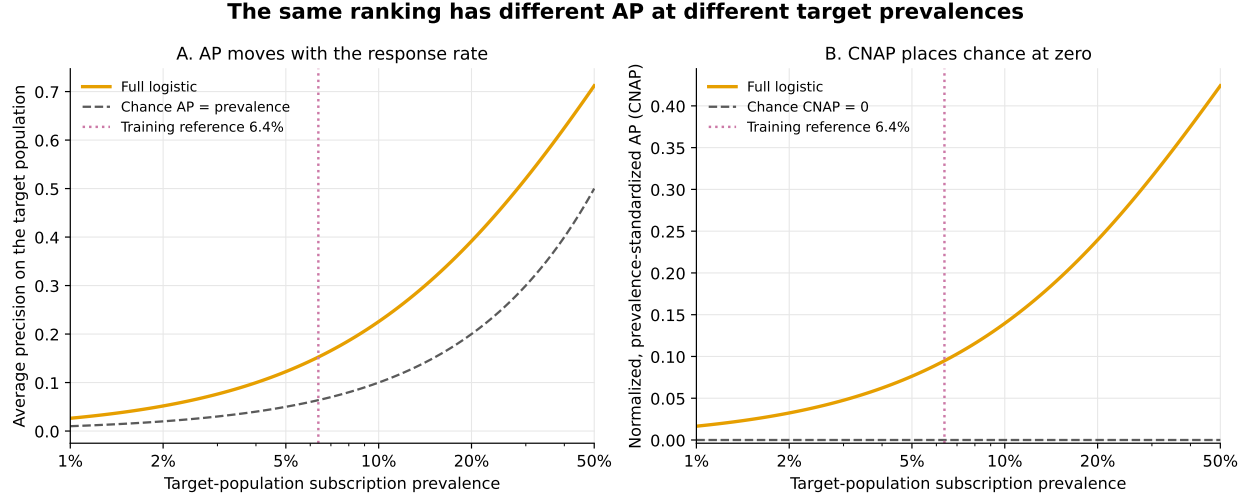


Figure 1: Prior-standardized AP and CNAP for one fitted ranking across the declared target-prevalence interval. The vertical line gives the training prevalence as a reference, not as a preferred target. Varying prevalence holds the observed class-conditional ranking behavior fixed.

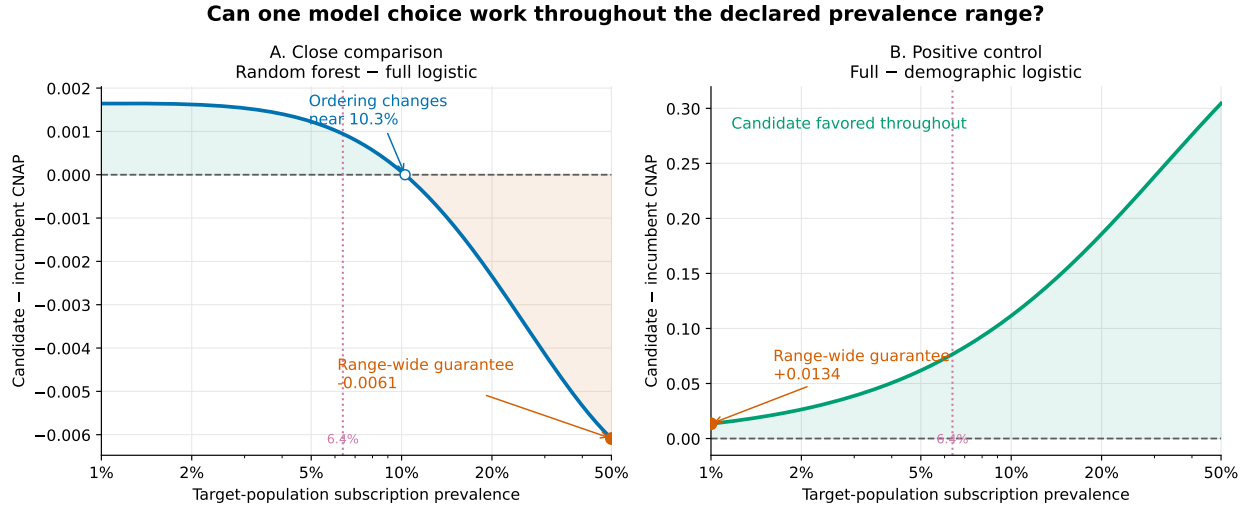


Figure 2: Paired CNAP differences in the single temporal evaluation. Each orange point is the infimum over the complete declared prevalence interval. The limiting prevalence is allowed to differ between comparisons.

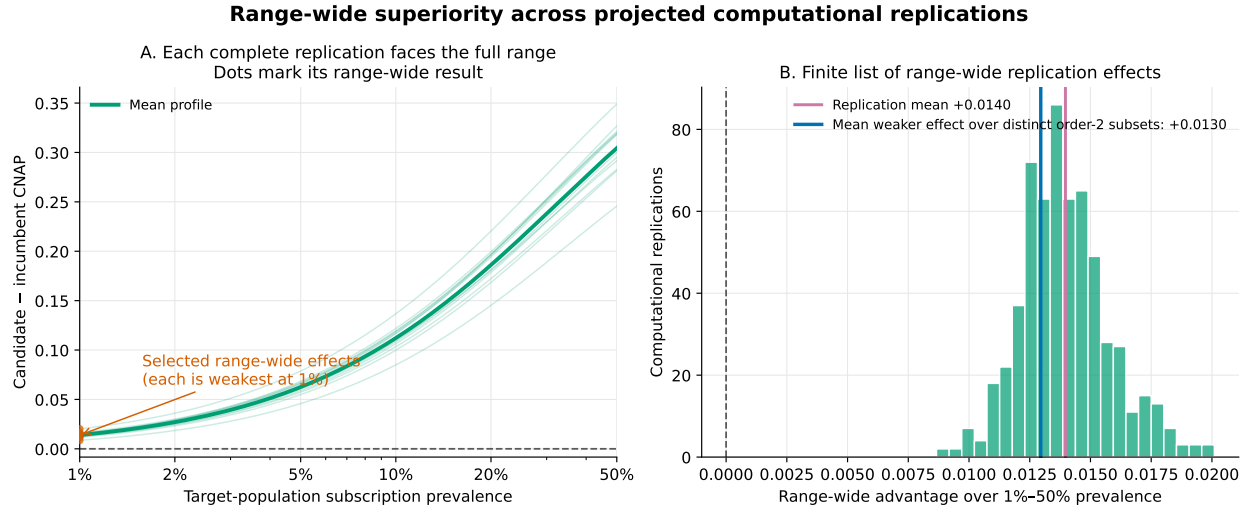


Figure 3: Projected support from one observed evaluation. Selected curves in panel A are complete paired computational replications; each dot is that replication’s retained effect. Panel B collects all 600 retained effects and contrasts their mean with order-2 supported magnitude. These replications do not constitute additional observed evaluations.

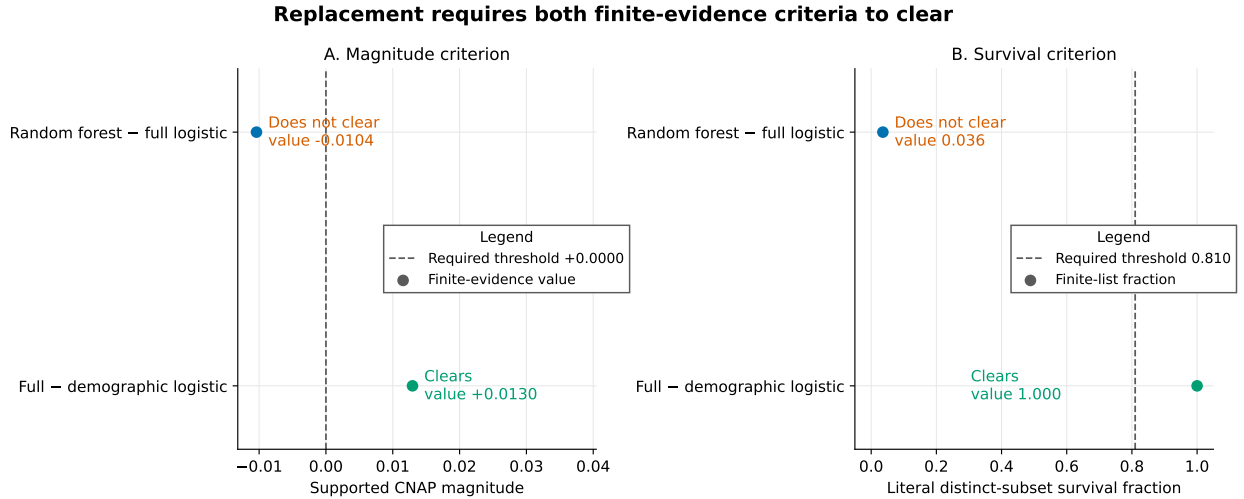


Figure 4: The two-part projected decision in the Bank Marketing illustration. Replacement requires both supported magnitude to exceed $\delta = 0$ and literal order-2 survival to exceed $\gamma = 0.81$. Failure returns no verdict, not a finding that the incumbent is superior.

The same fitted score pools also illustrate the AUROC extension in Appendix F. Pure prior change supplies no AUROC challenge, so the companion calculation increases the allowed within-class concentration on represented cases. Random forest against full logistic regression failed the policy at the observed mix. The positive control passed at $\Gamma = 1$ and first failed in the numerical interval $1.10 < \Gamma^\dagger \leq 1.20$, as shown in Figure 5. The interval is a numerical bracket, not a confidence interval.

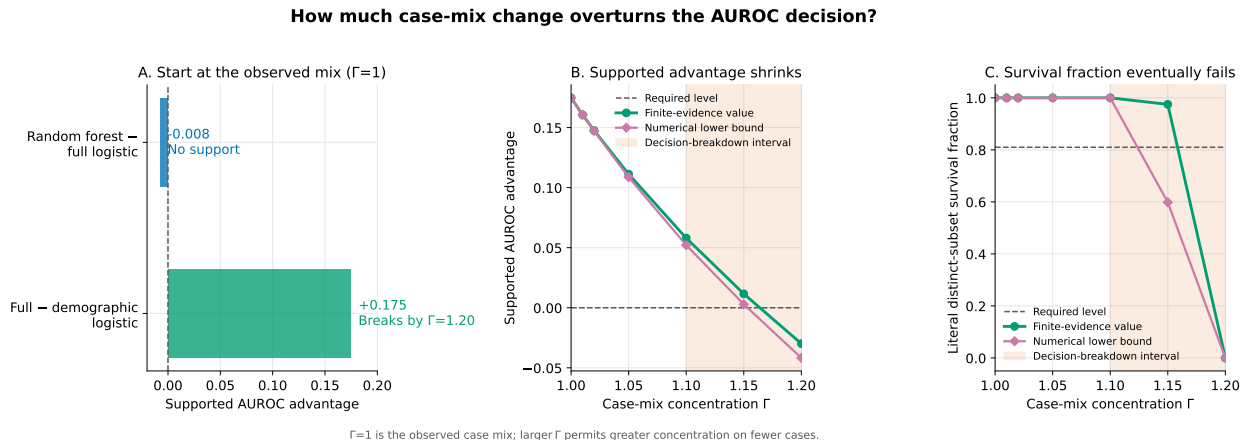


Figure 5: Projected empirical case-mix breakdown for the banking score pools. Larger Γ permits greater concentration on fewer represented cases. The challenge cannot introduce customer types absent from the temporal evaluation.

6.2 Several observed evaluations: ProteinGym

ProteinGym provides standardized deep-mutational-scanning assays together with released model scores [14]. We used version 1.3 substitution data, retained assays with manual biological cutoffs, and retained single amino-acid substitutions. Experimentally nonfunctional or deleterious variants formed the positive class. Because the released scores are oriented toward fitness, they were negated before analysis so that larger values favored the positive class. The three primary paired comparisons were ESM-1v ensemble against EVE ensemble, ESM-2 650M against ESM-1v ensemble, and ESM-2 650M against EVE ensemble. These scores come from already released models; no model was fitted or tuned for this study [8, 12, 11].

The aligned cohort contained 277,873 single substitutions in 91 assays, including 96,987 deleterious and 180,886 functional variants. Every retained variant had finite predictions from all compared models. Each assay supplied one full evaluation effect: the paired CNAP difference was computed throughout $[0.01, 0.50]$, and its infimum was retained. No computational resampling was performed. The resulting $M = 91$ effects therefore enter the observed path directly.

Figure 6 shows that the retained effects are heterogeneous for all three comparisons. The plots order assays separately within each comparison; horizontal position does not match the same assay between panels.

Here an order-2 pair means two complete assay evaluations, not two protein variants. There are $\binom{91}{2} = 4,095$ distinct assay pairs, and a pair survives at $d = 0$ only when both assays have positive retained effects. The strict requirement $\psi_{2,91,0} > 0.81$ therefore requires at least 82 of 91 assays to

Advantage retained across 1%–50% deleterious prevalence

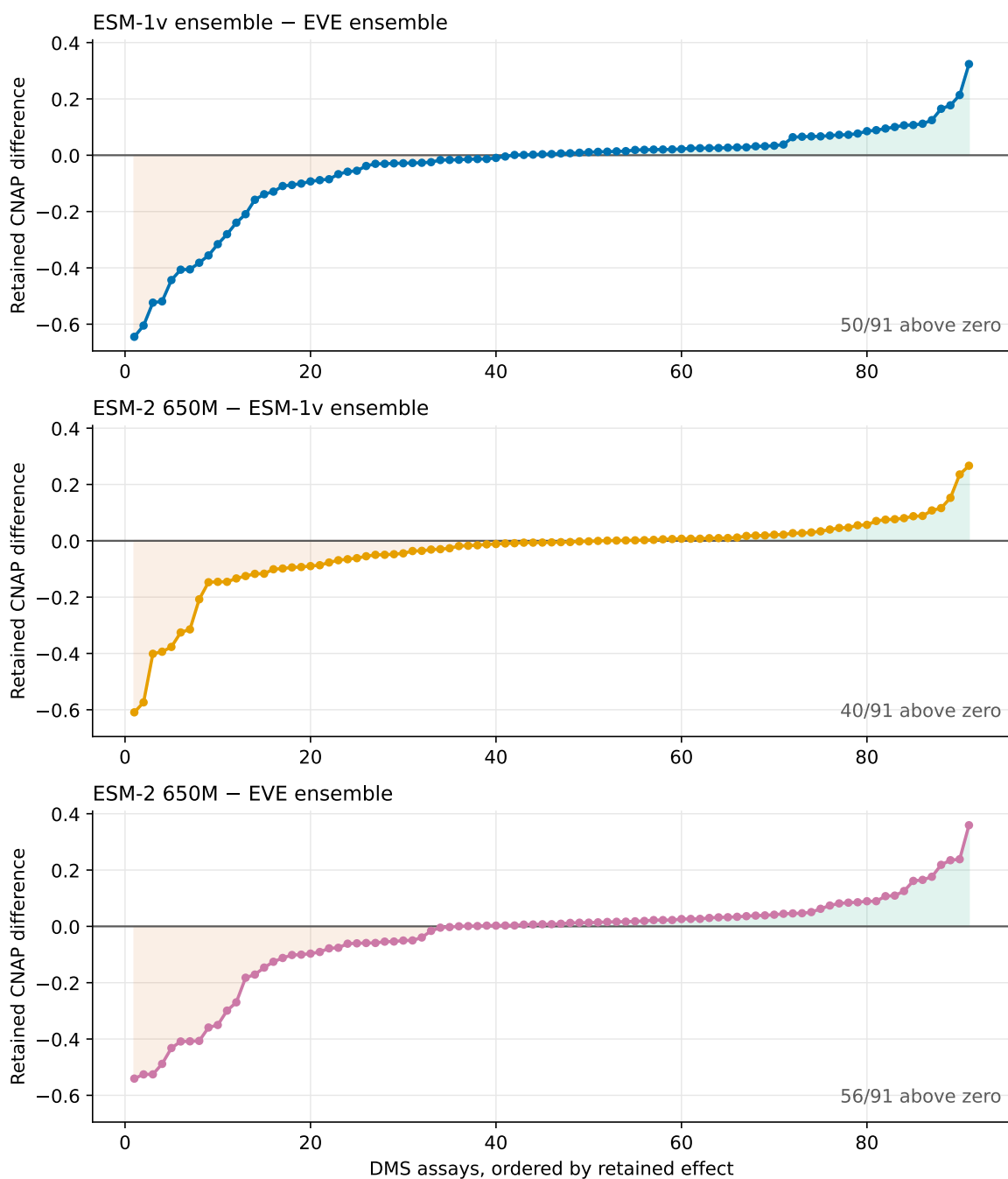


Figure 6: Observed retained assay effects for the three primary ProteinGym comparisons. Each point is one complete DMS assay after its paired model difference has faced the full declared prevalence interval. Green and orange regions distinguish positive from nonpositive retained effects; they are not uncertainty bands.

Table 1: Observed order-2 ProteinGym results. All comparisons use the same 91 assays and the same declared policy.

Candidate minus incumbent	Support	Assays > 0	Pair survival	Verdict
ESM-1v ensemble minus EVE ensemble	−0.1292	50/91	0.299	No verdict
ESM-2 650M minus ESM-1v ensemble	−0.1048	40/91	0.190	No verdict
ESM-2 650M minus EVE ensemble	−0.1264	56/91	0.376	No verdict

clear zero, since $\binom{82}{2}/\binom{91}{2} = 0.811 > 0.81$. None of the comparisons approached that count. ESM-2 exceeded EVE in 56 assays, for example, but the remaining losses were sufficiently severe that both mean retained effect and order-2 support were negative. Thus a majority win count did not support replacement.

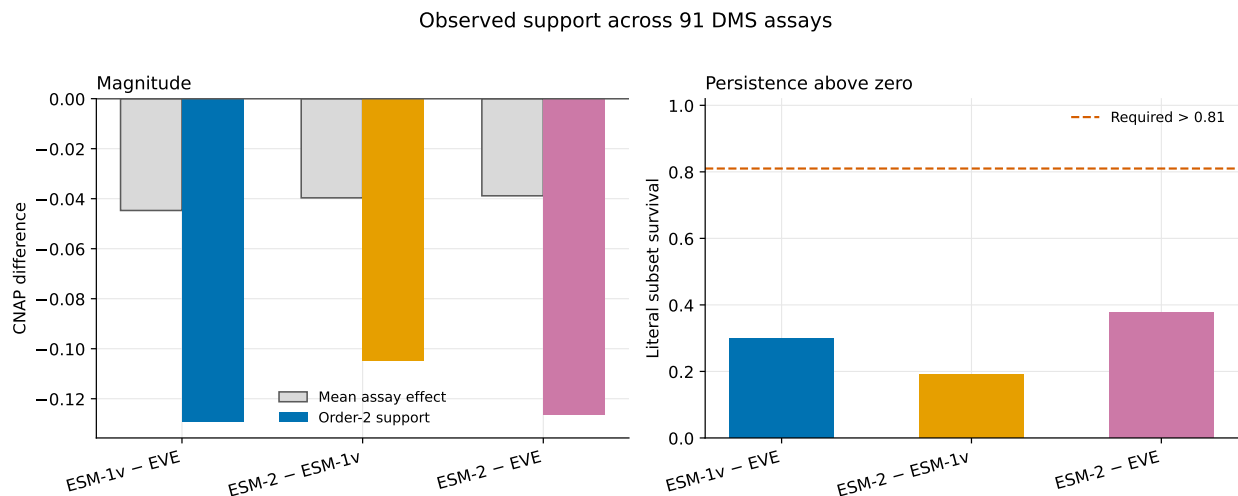


Figure 7: Observed supported magnitude and literal assay-pair survival for the three primary ProteinGym comparisons. The right panel’s bars are exact fractions of the 4,095 observed assay pairs whose two retained effects both exceed zero.

All three forward comparisons and all three reversed comparisons returned no verdict. The result does not establish a reverse hierarchy. It shows that the three released models do not form a stable replacement ordering across these assay contexts under the declared prevalence and order-2 challenges.

The evaluation regime treats a DMS assay as the full evaluation unit. The 91 assays represent 80 UniProt proteins and 60 publications; 20 assay rows share a protein with another retained assay, and 42 come from publications that contribute more than one assay. Those multiplicities are retained and reported rather than relabeled as independent research programs. Consequently, the claim is limited to the finite assay-level collection. The present primary analysis does not reduce the collection to one assay per publication.

7 Interpretation, reporting, and scope

7.1 What the quantities support

Supported magnitude and literal survival depend on the declared assessment and on the evaluations supplied to it; neither is a portable property of either model. They describe the same retained paired effects. Supported magnitude averages the weakest effect over subsets of K distinct evaluation effects, while literal survival records the fraction of those subsets whose effects all clear d . In the AP application, $\mathcal{C} = \Pi_*$ and the effect is the tie-averaged CNAP difference. Omitting any of those declarations changes what the quantities mean.

Projected and observed summaries require different language. Several separately conducted empirical replications give observed support across the tests satisfying the declared regime. A one-evaluation empirical-resampling procedure gives projected magnitude and survival by recombining the observed within-class score pools and should not be described as having survived external replications.

The result also remains a ranking-performance claim. A deployment decision may turn on calibration or on the consequences of acting on the selected cases, neither of which Equation (18) evaluates.

7.2 Relation to population-based robustness and inference

Several constructions in this paper have analogues in literatures that begin with a population probability law. The shared algebra does not imply a shared interpretation. Here support acts directly on subsets of a finite indexed list of retained effects. It neither posits nor estimates a superpopulation law for unseen evaluations. Probability enters only when a declared target population or deliberately constructed computational experiment requires it.

The retained effect $z_\ell = \inf_{c \in \mathcal{C}} \Delta_c(D_\ell)$ resembles an intersection bound. Chernozhukov, Lee, and Rosen study infima and suprema of population-indexed functions and develop bias correction and repeated-sampling inference for their bounds [5]. Equation (7) instead extracts the exact uniform lower level of the finite condition profile computed in evaluation D_ℓ . It neither estimates a functional $\inf_c \theta_c(P)$ nor receives a confidence interpretation from intersection-bound asymptotics.

Replication-survival support also has a finite spectral form. Let $\ell_1, \dots, \ell_L$ be the losses $-z_1, \dots, -z_L$,

with ordered values $\ell_{(1)} \leq \dots \leq \ell_{(L)}$. Then

$$-S_K(\mathbf{z}) = \sum_{r=K}^L \frac{\binom{r-1}{K-1}}{\binom{L}{K}} \ell_{(r)}. \quad (21)$$

The nondecreasing weights give $-S_K$ a finite spectral analogy [1], but they depend on both L and K . The manuscript invokes neither the associated coherence axioms nor a population-risk interpretation. Equation (11) is likewise related algebraically to distortion and Choquet representations of survival functions [21]. In those literatures the distribution is ordinarily the law of a random quantity and empirical order statistics estimate a population functional. Here Equations (21) and (27) are exact representations of a finite indexed list.

The capped AUROC weight sets in Appendix F are similarly related to distributionally robust optimization, which optimizes over alternative probability measures near a reference measure [19]. Ambiguity sets in that literature may be selected through divergence geometry and calibrated as confidence regions for uncertain probabilities [4]. The present cap instead declares how strongly the represented positive and negative cases may be redistributed. The factor Γ is a deterministic concentration allowance, not a confidence radius, a model of population shift, or a claim that an unknown true distribution lies in the admissible set.

Consequently, population consistency, asymptotic normality, and coverage results from these adjacent literatures are not invoked. Data-driven robust optimization can obtain such coverage by assuming independent observations from an unknown population law [9]; that is a different inferential target. Randomness deliberately introduced by the computational procedure in Section 5 supports Monte Carlo statements only about approximation of the declared projection. Additional empirical reach requires additional evaluations within $\mathcal{R}$, not a stronger probabilistic interpretation of the evaluations already observed.

7.3 Minimum reporting content

A concise report should identify the claim before presenting diagnostics. Table 2 gives the minimum content.

A possible headline statement for one observed evaluation is:

Under the declared assessment, model A was compared with model B using paired tie-averaged CNAP throughout Π_* . The analysis used J computational replications from one observed evaluation and reports projected support. Supported magnitude was [value], and literal survival above $d = [\text{value}]$ was [value]. The reversed orientation was [values]. Monte Carlo error was [value], and the prevalence search used tolerance [value]. With $\delta = [\text{value}]$ and $\gamma = [\text{value}]$, the finite-evidence rule returned [supported replacement/no verdict] relative to the observed evaluation and declared computational projection.

With several empirical evaluations, the second sentence should state, “The analysis used M separately conducted full evaluations satisfying the declared regime and reports observed support.” The report should give the individual contexts and the number of evaluations that cleared d .

Table 2: Minimum reporting content for a supported replacement analysis.

Component	Report
Claim and decision	Model orientation; observed $\theta_{A:B;K,M}^{\text{obs}}$ and $\psi_{K,M,d}^{A:B,\text{obs}}$, or projected $\hat{\theta}_{A:B;K,J}^{\text{proj}}$ and $\hat{\psi}_{K,J,d}^{A:B,\text{proj}}$; the reversed orientation; δ , d , and γ ; and the resulting finite-evidence verdict
Assessment scope	The admissibility requirements in $\mathcal{R}$; $\mathcal{C}$ and its scientific basis, with $\mathcal{C} = \Pi_\star$ for AP; what is fixed or may differ; the independent evaluation unit; and K
Evidence	The separately conducted empirical evaluations supplying the evidence; whether a computational projection was subsequently constructed; N or the evaluation-specific N_m ; M or J ; confirmation that the list length is at least K ; observed class counts; evaluation contexts; the number clearing d and the resulting fraction of distinct order- K subsets; and the empirical-resampling procedure where applicable
Effect construction	The invariance used to evaluate every $c \in \mathcal{C}$ from one evaluation; paired evaluation and resampling; the tie-averaged convention; and whether the diagnostic exchangeable-label randomization applies, with its scientific basis
Computation and diagnostics	Condition set or optimizer; numerical tolerance; computational-replication count J , Monte Carlo error, seed stability, and convergence where applicable; paired-effect profiles and minimizing conditions; evaluation effects; values near d ; and sensitivity to K , $\mathcal{C}$, and prespecified alternative floors

7.4 Limits of the AP claim

Prior standardization does not transport ranking behavior across changes within outcome class. If severity, measurement, or feature distributions change $P(S \mid Y)$, target-population evaluations are required.

Tie averaging does not create full label exchangeability. In matched, clustered, or heterogeneous-risk designs, the paired quantity remains computable but may inherit a design-specific anchor unless the admissible comparisons respect the design’s symmetry.

The choices of $\Pi_\star$, $\mathcal{R}$, and K remain scientific commitments. Widening $\Pi_\star$ or increasing K mechanically lowers the criterion, but neither operation proves that the resulting assessment captures the failures relevant to deployment.

7.5 Beyond prevalence: empirical case-mix breakdown for AUROC

The support construction also applies beyond prevalence and AP. Pure prior shift leaves AUROC unchanged, while changes in the mix of cases within each outcome class can challenge an AUROC advantage. Appendix F defines an empirical concentration challenge and the case-mix breakdown factor at which the replacement policy first fails. The result remains limited to the case types represented in the evaluation.

8 Conclusion

A model-replacement claim based on average precision must survive more than a favorable marginal summary. The population must be named because AP changes with prevalence, while the evaluations must be named because an average advantage can conceal a failure to persist. These are instances of the same noncompensation requirement, but they enter the assessment differently: target prevalence can be evaluated within each replication, whereas other variation must be represented by separate evaluations admitted by $\mathcal{R}$.

Prior standardization evaluates each model at the target prevalences when class-conditional score behavior is invariant. Chance normalization places those values on a common population random-to-perfect scale. Tie averaging then removes expected credit for arbitrary refinement before the models are compared on the same units. For J computational replications from one observed evaluation, the prevalence infimum is applied within each replication before projected order- K support:

$$\hat{\theta}_{A:B;K,J}^{\text{proj}} = \left(\frac{J}{K} \right)^{-1} \sum_{\substack{I \subseteq \{1, \dots, J\} \\ |I|=K}} \min_{j \in I} \inf_{\pi \in \Pi_\star} \{ \widetilde{\text{CNAP}}_{A,\pi}(D^{(j)}) - \widetilde{\text{CNAP}}_{B,\pi}(D^{(j)}) \}.$$

For M separately conducted empirical evaluations, observed support has the same form with M and D_m .

Supported magnitude and empirical literal survival summarize different features of the same challenged advantage. The finite-evidence replacement rule requires retained magnitude to exceed δ and the fraction of distinct order- K subsets whose effects all clear floor d to exceed γ . The first requirement preserves practical magnitude and sensitivity to severe losses; the second prevents favorable outcomes from compensating for excessive failure at the declared floor. Passing the rule supports replacement relative to the observed evaluations and declared challenges, not as a confidence statement about a latent population quantity.

Several empirical replications give both quantities from the same evaluation effects. One evaluation can support only empirical-resampling projections under a declared computational procedure. Those projections can guide a replacement analysis, but they cannot substitute computational resampling for new empirical opportunities to reveal failure.

The empirical illustrations show the distinction directly. In the banking study, the positive control clears both projected requirements relative to one temporal evaluation, while the close comparison returns no verdict. In the ProteinGym study, the observed assay list supplies genuine contextual heterogeneity: all three model comparisons return no verdict even when one candidate has a positive retained effect in a majority of assays. Win counts and marginal averages therefore do not replace the magnitude and persistence questions posed by the support operator.

The condition-indexed support operator also applies beyond prevalence. For AUROC, a symmetric capped concentration challenge can search over increasingly adverse redistributions of the observed positive and negative cases and report the empirical case-mix breakdown factor at which the same replacement policy first fails. This extension supplies a reproducible empirical robustness assessment when no target mix is selected. The search rule also removes the need to choose one concentration bound for the primary analysis. The assessment remains silent about case types absent from the evaluation.

A Empirical prior-standardized average precision

A.1 Definition and weighted implementation

Let

$$D = \{(s_i, y_i)\}_{i=1}^N, \quad y_i \in \{0, 1\},$$

where N is the number of evaluation units on which AP is calculated. Let $N_+ = \sum_i y_i > 0$ and $N_- = N - N_+ > 0$ be the positive and negative counts. Let $c_1 > \dots > c_H$ be the distinct observed score values. At threshold h , define

$$\text{TP}_h = \sum_i \mathbf{1}\{y_i = 1, s_i \geq c_h\}, \quad \text{FP}_h = \sum_i \mathbf{1}\{y_i = 0, s_i \geq c_h\},$$

and

$$r_h = \frac{\text{TP}_h}{N_+}, \quad f_h = \frac{\text{FP}_h}{N_-}.$$

With $r_0 = 0$, noninterpolated prior-standardized AP under the threshold-block convention is

$$\text{AP}_\pi(D) = \sum_{h=1}^H (r_h - r_{h-1}) \frac{\pi r_h}{\pi r_h + (1 - \pi) f_h}. \quad (22)$$

All units sharing a score enter one threshold block. A block containing only negative cases has zero recall increment, although its false positives reduce the precision of later blocks.

The same quantity can be computed by weighting each positive unit by $w_+ = \pi/N_+$ and each negative unit by $w_- = (1 - \pi)/N_-$. Weighted precision at threshold h is Equation (2), while weighted recall remains r_h .

A.2 Behavior across prevalence

Combining Equations (4) and (22) gives

$$\text{CNAP}_\pi(D) = \sum_{h=1}^H (r_h - r_{h-1}) \frac{\pi(r_h - f_h)}{\pi r_h + (1 - \pi) f_h}. \quad (23)$$

The derivative of a threshold contribution is

$$\frac{\partial}{\partial \pi} \left\{ \frac{\pi(r_h - f_h)}{\pi r_h + (1 - \pi) f_h} \right\} = \frac{f_h(r_h - f_h)}{\{\pi r_h + (1 - \pi) f_h\}^2}. \quad (24)$$

If $r_h \geq f_h$ at every threshold with positive recall increment, the profile is nondecreasing and its minimum over $[\pi_L, \pi_U]$ occurs at π_L . A paired difference can remain nonmonotone even when both marginal profiles are nondecreasing.

B Properties of replication-survival support

B.1 Survival and ordered representations

For a finite list $\mathbf{t} = (t_1, \dots, t_L)$ with entries in $[a, b]$, let

$$R_s = \sum_{\ell=1}^L \mathbf{1}\{t_\ell > s\}.$$

For each $I \in \mathcal{I}_{L,K}$,

$$\min_{\ell \in I} t_\ell = a + \int_a^b \mathbf{1}\{t_\ell > s \text{ for every } \ell \in I\} ds.$$

Among the $\binom{L}{K}$ subsets, exactly $\binom{R_s}{K}$ have every effect above s . Averaging the preceding identity over the subsets gives

$$S_K(\mathbf{t}) = a + \int_a^b \frac{\binom{R_s}{K}}{\binom{L}{K}} ds. \quad (25)$$

At each level s , the integrand is the fraction of distinct order- K subsets for which every effect retains the level.

B.2 Literal survival of a practical margin

For a prespecified practical margin d , define

$$\psi_{K,d}(\mathbf{t}) = \frac{\binom{R_d}{K}}{\binom{L}{K}}. \quad (26)$$

This quantity is a finite-list subset fraction at one effect level. Equation (25) can be written

$$S_K(\mathbf{t}) = a + \int_a^b \psi_{K,s}(\mathbf{t}) ds,$$

so replication-survival support integrates finite-list literal survival over all effect levels.

For an observed effect list, literal survival is an exact fraction of its distinct subsets. For a projected list, finite J introduces only Monte Carlo approximation to the declared empirical-resampling projection. Because AP is discrete, the choice between $>$ and $\geq$ and sensitivity to values near d must be declared.

Let $t_{(1)} \leq \dots \leq t_{(L)}$. The minimum of a distinct order- K subset equals $t_{(i)}$ for exactly $\binom{L-i}{K-1}$ subsets. Hence

$$S_K(\mathbf{t}) = \binom{L}{K}^{-1} \sum_{i=1}^{L-K+1} \binom{L-i}{K-1} t_{(i)}. \quad (27)$$

The weights move toward the lower observed effects as K increases. This exact finite-list representation has the spectral and distortion analogues discussed in Section 7.2 [1, 21].

B.3 Proof of the operator-order inequality

For each $I \in \mathcal{I}_{L,K}$, let $X_c^{(\ell)} = \Delta_c(D_\ell)$ and $Y_{c,I} = \min_{\ell \in I} X_c^{(\ell)}$. Then

$$\min_{\ell \in I} \inf_{c \in \mathcal{C}} X_c^{(\ell)} = \inf_{c \in \mathcal{C}} \min_{\ell \in I} X_c^{(\ell)} = \inf_{c \in \mathcal{C}} Y_{c,I}.$$

Since $\inf_c Y_{c,I} \leq Y_{c',I}$ for every $c' \in \mathcal{C}$, averaging over $I \in \mathcal{I}_{L,K}$ gives

$$\binom{L}{K}^{-1} \sum_{I \in \mathcal{I}_{L,K}} \inf_{c \in \mathcal{C}} Y_{c,I} \leq \inf_{c \in \mathcal{C}} \binom{L}{K}^{-1} \sum_{I \in \mathcal{I}_{L,K}} Y_{c,I},$$

which proves Equation (12).

B.4 Proof of challenge monotonicity

Take $\mathcal{C}_1 \subseteq \mathcal{C}_2$ and $1 \leq K_1 \leq K_2 \leq L$. Expanding the condition set lowers each evaluation effect, so it remains to compare the subset orders for one fixed list. Every order- K_2 subset contains $\binom{K_2}{K_1}$ order- K_1 subsets, and its minimum is no greater than the average minimum of those subsets. Averaging this inequality over all order- K_2 subsets counts each order- K_1 subset exactly $\binom{L-K_1}{K_2-K_1}$ times. The resulting averages are S_{K_2} and S_{K_1} , which proves Proposition 1.

C Paired comparison proofs and counterexamples

C.1 Why marginal subtraction overstates paired advantage

Let U_π and V_π be two profiles, with

$$U_{\Pi_\star} = \inf_{\pi \in \Pi_\star} U_\pi, \quad V_{\Pi_\star} = \inf_{\pi \in \Pi_\star} V_\pi, \quad T_{\Pi_\star} = \inf_{\pi \in \Pi_\star} (U_\pi - V_\pi).$$

Choosing a prevalence minimizing U_π gives

$$T_{\Pi_\star} \leq U_{\Pi_\star} - V_\pi \leq U_{\Pi_\star} - V_{\Pi_\star}.$$

Monotonicity of S_K therefore gives

$$S_K(U_{\Pi_\star} - V_{\Pi_\star}) \geq S_K(T_{\Pi_\star}).$$

For effects evaluated on the same replications, the minimum is superadditive:

$$\min_j (X_j + Y_j) \geq \min_j X_j + \min_j Y_j.$$

Taking $X = U_{\Pi_\star} - V_{\Pi_\star}$ and $Y = V_{\Pi_\star}$ yields

$$S_K(U_{\Pi_\star}) - S_K(V_{\Pi_\star}) \geq S_K(U_{\Pi_\star} - V_{\Pi_\star}),$$

which completes Equation (13).

C.2 Proof of refinement neutrality

Condition on S_B and Y . Proposition 2 makes model A 's strict ranking O_A uniform on $\mathcal{O}(D_B)$. Tie-averaged AP for A is the conditional expectation of ordinary AP over whatever ties A retains. By the tower property,

$$\mathbb{E}[\widetilde{\text{AP}}_{A,\pi} \mid S_B, Y] = \mathbb{E}[\text{AP}_\pi(D_{O_A}) \mid S_B, Y].$$

Uniformity of O_A turns the right side into the average of AP_π over $\mathcal{O}(D_B)$, which is $\widetilde{\text{AP}}_{B,\pi}$.

C.3 Rank invariance under exchangeable labels

Fix N_+ , N_- , and a strict ranking o . Under Definition 3, the label sequence read along o is uniform over the $\binom{N}{N_+}$ arrangements containing N_+ positives. Reading along another strict ranking is a bijection of the same arrangements, so the sequence has the same law.

For a strict ranking, $\pi \mapsto \text{CNAP}_\pi(D_o)$ is a deterministic function of that sequence. Its expectation is therefore the same for every strict ranking. Tie-averaged AP is a convex combination of strict-ranking values, so any two score vectors on the same units satisfy

$$\mathbb{E}_0[\widetilde{\text{CNAP}}_{A,\pi}] = \mathbb{E}_0[\widetilde{\text{CNAP}}_{B,\pi}].$$

Thus $\mathbb{E}_0[\Delta_\pi^{A:B}(D)] = 0$.

For a fixed evaluation D ,

$$\mathbb{E}_0 \left[\inf_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D) \right] \leq \inf_{\pi \in \Pi_\star} \mathbb{E}_0[\Delta_\pi^{A:B}(D)] = 0,$$

and $S_K(\mathbf{t}) \leq L^{-1} \sum_{\ell=1}^L t_\ell$ for any finite list of length L . Applying this inequality to every label randomization and then taking $\mathbb{E}_0$ proves Proposition 3; exchanging A and B proves the reversed orientation.

C.4 Why restricted label symmetry is insufficient

Take four units in two matched pairs, each containing exactly one positive. The design permits labels to swap within pairs but not between pairs. Let model A give all units the same score, while model B places the first pair above the second and ties the units within each pair. At $\pi = 1/2$, exact tie averaging over the four permitted assignments gives

$$\widetilde{\text{CNAP}}_A - \widetilde{\text{CNAP}}_B = \frac{1}{36}$$

in one orientation for every assignment. The effect has no replication variation, so its supported value remains $1/36$ for every admissible K .

The obstruction is the symmetry group of the design. Full exchangeability places every strict ranking in one orbit of the label-permutation group. Restricted designs may divide rankings into several orbits with different anchors, and averaging within score ties cannot move a ranking between them.

C.5 Why paired null centering fails

Take two positive and two negative units at $\pi = 1/2$, with

$$s_A = (4, 3, 2, 1), \quad s_B = (2, 2, 1, 1),$$

and let the two highest-scored units be positive. Both models separate the classes perfectly, so every stratified empirical-resampling evaluation has CNAP one for each model and paired difference zero.

Under joint label permutation, exact enumeration of the $K = 2$ support calculation gives an expected supported value of $35/384$ for $A - B$ and $-295/1152$ for $B - A$. Subtracting these reference values would report a positive corrected value of about 0.256 for the second orientation, favoring the coarser model although both predictions are perfect. The displacement is not an additive constant because perfect separation leaves no room for it.

C.6 Proof of the orientation bound

For evaluation j , write

$$L_j = \inf_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D_j), \quad U_j = \sup_{\pi \in \Pi_\star} \Delta_\pi^{A:B}(D_j).$$

For each $I \in \mathcal{I}_{L,K}$, the two orientations contribute

$$\min_{j \in I} L_j \quad \text{and} \quad -\max_{j \in I} U_j.$$

Since $L_j \leq U_j$ for every evaluation,

$$\min_{j \in I} L_j - \max_{j \in I} U_j \leq 0,$$

and averaging over the distinct order- K subsets proves Equation (16). The negative of this sum is the width of the no-verdict region. When $\Pi_\star$ is a singleton and $K = 2$, the width is

$$\binom{L}{2}^{-1} \sum_{1 \leq i < j \leq L} |T_i - T_j|$$

for the finite paired-effect list.

D Closed-form computation of tie-averaged AP

Index the H distinct score blocks by $h = 1, \dots, H$ from highest to lowest. Let block h contain a_h positive and b_h negative units, with $m_h = a_h + b_h$, and define

$$A_h = \sum_{\ell < h} a_\ell, \quad B_h = \sum_{\ell < h} b_\ell.$$

Then

$$\widetilde{\text{AP}}_\pi(D) = \sum_{h=1}^H \frac{a_h}{N_+} \frac{1}{m_h} \sum_{k=1}^{m_h} \sum_{t=\max(0, k-1-b_h)}^{\min(k-1, a_h-1)} \frac{\binom{a_h-1}{t} \binom{b_h}{k-1-t}}{\binom{m_h-1}{k-1}} \frac{\pi r_{hkt}}{\pi r_{hkt} + (1-\pi) f_{hkt}}, \quad (28)$$

where

$$r_{hkt} = \frac{A_h + t + 1}{N_+}, \quad f_{hkt} = \frac{B_h + k - 1 - t}{N_-}.$$

To obtain the formula, fix one positive unit in block h . Under a uniformly random ordering of the block, its position k is uniform on $\{1, \dots, m_h\}$. Given k , the number t of other positive units in the $k-1$ preceding positions is hypergeometric with a_h-1 available positives and b_h negatives. The final fraction in Equation (28) is the prior-standardized precision when that positive enters the ranking. Averaging over t and k , multiplying by the block's a_h positive units, and summing over blocks gives the result.

The calculation costs $O(\sum_h m_h^2)$ operations. Its combinatorial weights do not depend on π and can be reused throughout the prevalence search. When all scores are distinct, Equation (28) reduces to ordinary stepwise AP.

E Computation and approximation

E.1 Several empirical evaluations

Suppose M separately conducted full evaluations $D_1, \dots, D_M$ satisfy $\mathcal{R}$. Let N_m be the number of evaluation units in D_m . The values $N_1, \dots, N_M$ may differ, and the evaluations may occur in distinct admissible contexts. For each evaluation, compute

$$z_m = \inf_{c \in \mathcal{C}} \Delta_c(D_m).$$

Let $\mathbf{z}_M^{\text{obs}} = (z_1, \dots, z_M)$. When $K \leq M$, observed support is

$$S_{K,M,\mathcal{C}}^{\text{obs}} = S_K(\mathbf{z}_M^{\text{obs}}) = \binom{M}{K}^{-1} \sum_{I \in \mathcal{I}_{M,K}} \min_{m \in I} z_m. \quad (29)$$

For AP replacement, $\mathcal{C} = \Pi_\star$ and the effect profile is $\Delta_\pi^{A:B}$, giving $\theta_{A:B;K,M}^{\text{obs}}$. The same effects give

$$\psi_{K,M,d}^{A:B,\text{obs}} = \frac{\binom{R_d}{K}}{\binom{M}{K}}, \quad R_d = \sum_{m=1}^M \mathbf{1}\{z_m > d\}. \quad (30)$$

These are exact functionals of the finite indexed effect list. Each distinct evaluation receives equal influence in the subset construction. A small M gives few distinct subsets. Increasing K shifts support toward the lower effects and requires more effects to clear d jointly. If $M < K$, the declared observed assessment is unavailable. The count M records the number of observed evaluations.

If the fitted models change between evaluations, the evaluations address one scoped claim only when a common prespecified procedure produces those models. The count M includes only evaluations whose model construction was not altered in response to another evaluation in the set. Updates prompted by an earlier evaluation create an adaptive sequence that requires a separate ordered analysis. Dataset partitions with no scientific identity should not be used to increase M .

E.2 Sorted computation

Sort the M observed evaluation effects:

$$z_{(1)} \leq \dots \leq z_{(M)}.$$

Exactly $\binom{M-i}{K-1}$ distinct order- K subsets have minimum $z_{(i)}$. Hence Equation (29) equals

$$S_{K,M,\mathcal{C}}^{\text{obs}} = \binom{M}{K}^{-1} \sum_{i=1}^{M-K+1} \binom{M-i}{K-1} z_{(i)}. \quad (31)$$

For J empirical-resampling effects, the Monte Carlo approximation is

$$\hat{S}_{K,J,\mathcal{C}}^{\text{proj}} = \binom{J}{K}^{-1} \sum_{i=1}^{J-K+1} \binom{J-i}{K-1} z_{(i)}. \quad (32)$$

At $K = 2$, this reduces to

$$\hat{S}_{2,J,\mathcal{C}}^{\text{proj}} = \frac{2}{J(J-1)} \sum_{i=1}^{J-1} (J-i) z_{(i)}.$$

E.3 Monte Carlo error

For $K = 2$, define the leave-one-replication projection

$$\hat{h}_{1,-i}(z_i) = \frac{1}{J-1} \sum_{\substack{j=1 \\ j \neq i}}^J \min(z_i, z_j).$$

A first-order Monte Carlo standard error for the projected approximation is

$$\widehat{\text{se}}_{\text{MC}} = \frac{2}{\sqrt{J}} \text{sd}_i \left\{ \hat{h}_{1,-i}(z_i) \right\}. \quad (33)$$

Prefix sums over the sorted array compute all projections in linear time after sorting.

E.4 Prevalence evaluation and numerical checks

For a finite prevalence set with G values, score-block counts are computed once in each evaluation and reused across prevalences. For a continuous interval, Equation (23) is a smooth one-dimensional objective away from thresholds with no recall contribution. Endpoint values and stationary points may be checked directly, while a numerical optimizer should be validated against a dense grid. The report should give the optimizer tolerance and should distinguish stability of the minimum value from stability of the minimizing prevalence. For projected support, calculations across larger values of J and independent random seeds assess Monte Carlo convergence to the declared empirical-resampling projection.

F Empirical case-mix breakdown for AUROC

F.1 Prior shift supplies no AUROC challenge

The condition set used for AP is a set of target prevalences. It is available because AP depends on the class prior. AUROC does not. AUROC is the probability that a randomly drawn positive outranks a randomly drawn negative, with half credit for a tie, and is therefore a functional of the two class-conditional score distributions alone. Changing the prior alters how often each class is drawn while leaving both conditionals intact. An AUROC advantage that survives every $\pi \in \Pi_\star$ has consequently survived no additional challenge.

The within-class composition remains free to change, however. A new population may contain different proportions of mild and severe positives or easy and difficult negatives even when its prevalence is unchanged. Named target mixes can be evaluated when those proportions and the required transport assumptions are available. For settings without named target mixes, the observed cases define the case support for the challenge. The procedure admits increasingly concentrated redistributions within each outcome class and reports the first concentration at which either replacement criterion fails. It begins with uniform within-class weights and progressively relaxes a common upper bound on the weight any observation may receive. Reporting the breakdown factor removes the need to select one concentration bound for the primary analysis. The bound and the resulting breakdown factor are defined below.

F.2 The paired AUROC difference as a case-mix average

Let D contain paired scores (s_i^A, s_i^B) and outcome Y_i for every evaluation unit, with positive index set $\mathcal{I}_+$, negative index set $\mathcal{I}_-$, and class counts N_+ and N_- . Thus the dataset contains $N = N_+ + N_-$ evaluation units. For model $m \in \{A, B\}$, write

$$\phi_m(i, k) = \mathbf{1}\{s_i^m > s_k^m\} + \frac{1}{2}\mathbf{1}\{s_i^m = s_k^m\}, \quad i \in \mathcal{I}_+, \quad k \in \mathcal{I}_-.$$

Thus $\phi_m(i, k)$ is one when model m orders the positive-negative pair correctly, one half for a tie, and zero otherwise. Define the pairwise gain matrix

$$G_{ik} = \phi_A(i, k) - \phi_B(i, k).$$

If positive and negative cases receive normalized weights w_+ and w_- , the paired AUROC difference under that empirical case mix is

$$\Delta_D(w_+, w_-) = \sum_{i \in \mathcal{I}_+} \sum_{k \in \mathcal{I}_-} w_{+,i} w_{-,k} G_{ik}. \quad (34)$$

Uniform within-class weights recover the ordinary empirical AUROC difference. Because both models are evaluated with the same w_+ and w_- , the redistribution remains paired.

No additional tie correction is needed. Each pair already receives half credit for a tie, so exchangeable labels give every fixed score vector mean AUROC one half regardless of its tie structure. The positive finite-sample reference value that required tie-averaged AP in Section 4.3 does not recur.

F.3 Why unrestricted redistribution degenerates

If the weights in Equation (34) are otherwise unrestricted, all positive weight may be placed on one positive case and all negative weight on one negative case. The infimum then reduces to the least favorable single entry of G . This is too concentrated to support a useful replacement criterion.

The degeneracy is especially clear when candidate A separates perfectly. Then $G_{ik} = 1 - \phi_B(i, k) \geq 0$ for every pair, but concentrating on any pair that incumbent B also orders correctly gives difference zero. A perfect candidate can show a strictly positive worst-case advantage only if the incumbent fails on every pair. Unrestricted empirical reweighting therefore does not make replacement appropriately demanding; it generally makes a strict advantage unavailable.

F.4 The symmetric capped concentration challenge

What causes the degeneracy is concentration rather than empirical reweighting itself. For a common factor $\Gamma \geq 1$, define

$$\mathcal{W}_{+,\Gamma}(D) = \left\{ w_+ : \sum_{i \in \mathcal{I}_+} w_{+,i} = 1, \quad 0 \leq w_{+,i} \leq \frac{\Gamma}{N_+} \right\}, \quad (35)$$

and

$$\mathcal{W}_{-,\Gamma}(D) = \left\{ w_- : \sum_{k \in \mathcal{I}_-} w_{-,k} = 1, \quad 0 \leq w_{-,k} \leq \frac{\Gamma}{N_-} \right\}. \quad (36)$$

The declared condition set is

$$\mathcal{C}_\Gamma(D) = \mathcal{W}_{+, \Gamma}(D) \times \mathcal{W}_{-, \Gamma}(D).$$

This capped density family is a finite empirical analogue of an ambiguity set in distributionally robust optimization [19, 4]. Here it is fixed as a deterministic challenge over represented cases; it is not calibrated as a confidence region for an unknown distribution. The advantage retained in evaluation D is

$$Z_\Gamma^{A:B}(D) = \inf_{(w_+, w_-) \in \mathcal{C}_\Gamma(D)} \Delta_D(w_+, w_-). \quad (37)$$

At $\Gamma = 1$, normalization forces every weight to be uniform, so

$$Z_1^{A:B}(D) = \text{AUC}_A(D) - \text{AUC}_B(D). \quad (38)$$

At larger values, no observation may receive more than Γ times its uniform within-class weight. The bound is one-sided: an observation may receive zero weight, while concentration on the retained observations is limited from above. The effective sample size (ESS) summarizes this concentration. For either class y ,

$$\sum_i w_{y,i}^2 \leq \left(\max_i w_{y,i} \right) \sum_i w_{y,i} \leq \frac{\Gamma}{N_y},$$

and hence

$$\text{ESS}(w_y) = \left(\sum_i w_{y,i}^2 \right)^{-1} \geq \frac{N_y}{\Gamma}. \quad (39)$$

Thus $1/\Gamma$ is a guaranteed minimum effective-support fraction implied by the cap, although an effective-sample-size restriction alone would not define the same set of weights.

The admissible sets are nested:

$$\Gamma_1 \leq \Gamma_2 \implies \mathcal{C}_{\Gamma_1}(D) \subseteq \mathcal{C}_{\Gamma_2}(D),$$

so

$$Z_{\Gamma_2}^{A:B}(D) \leq Z_{\Gamma_1}^{A:B}(D). \quad (40)$$

A perfectly separating model continues to attain AUROC one throughout every admissible set. The challenge controls the maximum concentration allowed among the observed cases in each outcome class.

The common factor Γ gives both classes the same normalized upper allowance. Their realized redistributions remain separate because the positive and negative weights are optimized independently. Suppose separate bounds Γ_+ and Γ_- were introduced. Every combination satisfying $1 \leq \Gamma_+ \leq \Gamma$ and $1 \leq \Gamma_- \leq \Gamma$ would lie in a square of weaker concentration allowances. Monotonicity makes (Γ, Γ) its most adverse corner. Equation (37) therefore challenges that square with one concentration coordinate. Asymmetric bounds may be useful as diagnostics. The primary implementation and minimum report use the common factor.

F.5 Replication-challenged decision profiles

Apply the paper's support construction separately at every common concentration factor. For a finite list $\mathbf{D}$ of empirical or computational evaluations, let

$$\mathbf{z}(\Gamma; \mathbf{D}) = Z_\Gamma^{A:B}[\mathbf{D}].$$

Define the supported-magnitude profile

$$\theta_{A:B;K}(\Gamma; \mathbf{D}) = S_K(\mathbf{z}(\Gamma; \mathbf{D})) \quad (41)$$

and, for an AUROC-difference floor $d \geq 0$, the literal-survival profile

$$\psi_{K,d}^{A:B}(\Gamma; \mathbf{D}) = \psi_{K,d}(\mathbf{z}(\Gamma; \mathbf{D})). \quad (42)$$

The same common Γ is applied to every evaluation, while the adverse weights may differ between evaluations. Because Equation (40) holds pointwise, both $\theta_{A:B;K}(\Gamma; \mathbf{D})$ and $\psi_{K,d}^{A:B}(\Gamma; \mathbf{D})$ are nonincreasing in Γ .

Let $\delta \geq 0$ now denote the smallest supported AUROC advantage that would justify replacement, and retain $\gamma \in (0, 1)$ as the required literal survival fraction. At concentration factor Γ , the AUROC replacement policy requires

$$\theta_{A:B;K}(\Gamma; \mathbf{D}) > \delta \quad \text{and} \quad \psi_{K,d}^{A:B}(\Gamma; \mathbf{D}) > \gamma. \quad (43)$$

At $\Gamma = 1$, Equation (43) is the ordinary paired AUROC replacement assessment under the declared replication challenge. As Γ grows, it asks how far that same conclusion survives increasingly concentrated recombinations of the cases represented in each evaluation.

F.6 The empirical case-mix breakdown factor

The declared procedure locates the first value of Γ at which either the supported-magnitude requirement or the literal-survival requirement fails. The reported value is therefore fixed by the search rule.

Definition 4 (Empirical case-mix breakdown factor). *For the oriented comparison of candidate A with incumbent B , define*

$$\Gamma_{\text{ECM},A:B}^{\dagger}(\mathbf{D}) = \inf \left\{ \Gamma \geq 1 : \theta_{A:B;K}(\Gamma; \mathbf{D}) \leq \delta \text{ or } \psi_{K,d}^{A:B}(\Gamma; \mathbf{D}) \leq \gamma \right\}. \quad (44)$$

The quantity is the smallest common upper within-class concentration factor at which the replacement justification first fails. Equivalently, before the breakdown, the conclusion survives every pair of empirical positive- and negative-class redistributions whose individual weights are capped by Γ/N_+ and Γ/N_- . A larger value means that a stronger empirical concentration challenge is required to defeat the declared replacement policy.

If Equation (43) already fails at $\Gamma = 1$, then $\Gamma_{\text{ECM},A:B}^{\dagger}(\mathbf{D}) = 1$: there is no supported replacement conclusion before case-mix stress is applied. If the failure set in Equation (44) is empty, its infimum is $+\infty$ and the report should state that no breakdown occurred over the full range of redistributions permitted by the observed cases. In a finite evaluation the weight sets eventually saturate, because a cap of at least one permits unrestricted simplex weights. A zero crossing of $\theta_{A:B;K}(\Gamma; \mathbf{D})$ may be informative as a secondary sensitivity diagnostic, but the policy breakdown is primary because it preserves the paper’s practical-magnitude and literal-survival requirements.

The breakdown factor remains indexed by the declared assessment: the paired models, observed evaluations and case sets, evaluation regime, replication order, and policy thresholds δ , d , and γ . Its reproducibility requires the weight family and search rule to be fixed before the result is seen. The quantity remains dependent on the declared context.

F.7 Computation and breakdown reporting

Four counts have different roles. Let N_m be the number of evaluation units in evaluation D_m , let M be the number of separately conducted admissible empirical evaluations, let J be the number of computational empirical-resampling replications used when only one evaluation is observed, and let K be the declared order of the replication challenge. Increasing M adds empirical evidence, increasing J reduces Monte Carlo error in a projected analysis, and increasing K changes the criterion by placing greater emphasis on adverse evaluations. Observed support requires $M \geq K$, while projected support requires $J \geq K$; otherwise the corresponding order- K assessment is unavailable.

Several empirical evaluations. Suppose $D_1, \dots, D_M$ are separately conducted full evaluations satisfying $\mathcal{R}$. At each searched concentration factor, compute

$$z_m(\Gamma) = Z_{\Gamma}^{A:B}(D_m), \quad m = 1, \dots, M.$$

Applying the supported-magnitude and literal-survival summaries in Equations (29) and (30) directly to this indexed effect list gives

$$\theta_{A:B;K,M}^{\text{obs}}(\Gamma) \quad \text{and} \quad \psi_{K,M,d}^{A:B,\text{obs}}(\Gamma).$$

The observed-replication breakdown factor is

$$\Gamma_{\text{ECM},A:B}^{\dagger,\text{obs}} = \inf \left\{ \Gamma \geq 1 : \theta_{A:B;K,M}^{\text{obs}}(\Gamma) \leq \delta \text{ or } \psi_{K,M,d}^{A:B,\text{obs}}(\Gamma) \leq \gamma \right\}. \quad (45)$$

One observed evaluation. When only one observed full evaluation D_{obs} is available, generate J computational replications $D^{(1)}, \dots, D^{(J)}$ under the declared empirical-resampling procedure. Every computational replication must preserve the paired scores, follow that procedure, and repeat the optimization in Equation (37). For every computational replication and searched concentration factor, compute

$$z_j(\Gamma) = Z_{\Gamma}^{A:B}(D^{(j)}), \quad j = 1, \dots, J.$$

The same $D^{(j)}$ must be reused throughout the Γ search. Changes in $z_j(\Gamma)$ then arise from relaxing the concentration bound rather than from using different computational resamples at different values of Γ .

Applying the two order- K finite-list functionals to this indexed effect list gives

$$\hat{\theta}_{A:B;K,J}^{\text{proj}}(\Gamma) \quad \text{and} \quad \hat{\psi}_{K,J,d}^{A:B,\text{proj}}(\Gamma).$$

The first value of Γ at which either approximated profile crosses its policy threshold gives the projected empirical case-mix breakdown approximation

$$\hat{\Gamma}_{\text{ECM},A:B}^{\dagger,\text{proj}} = \inf \left\{ \Gamma \geq 1 : \hat{\theta}_{A:B;K,J}^{\text{proj}}(\Gamma) \leq \delta \text{ or } \hat{\psi}_{K,J,d}^{A:B,\text{proj}}(\Gamma) \leq \gamma \right\}. \quad (46)$$

The J computational replications describe the evaluation effects produced by the declared resampling procedure. They do not supply $M = J$ observed evaluations. Increasing J reduces Monte Carlo error but cannot expose a case type, environment, measurement process, or failure mechanism absent from D_{obs} . The resulting profiles and breakdown factor must therefore be described as projected.

For the observed path, the first failure point is a finite-list functional, subject only to numerical optimization error. For the projected path, finite J adds Monte Carlo approximation error under the constructed resampling law. Neither breakdown factor is an inferential guarantee about unseen cases or contexts. The projected calculation should be repeated across random seeds and larger values of J to assess stability of the first failure point.

F.8 Numerical verification and minimum reporting

Writing $G = (G_{ik})$, Equation (37) minimizes the bilinear form $w_{\perp}^{\top} G w_{\perp}$ over two capped simplices. A feasible weighting supplies an upper bound on the infimum and can demonstrate failure when its value violates a policy requirement. Numerically verifying that the requirement continues to hold needs a valid global lower bound; alternating optimization without such a bound may stop above the true infimum and overstate robustness.

Monotonicity permits a bracketed search for the corresponding observed or projected breakdown factor rather than evaluation of a dense curve solely for reporting. The numerical tolerance, global-optimization gap, Monte Carlo error, and seed and J stability should be recorded. The full profiles remain useful diagnostics and should be retained by the software, but they need not appear in the headline report.

A minimum report gives the decision results at $\Gamma = 1$, the computed breakdown factor, its numerical tolerance and optimization gap, Monte Carlo error where applicable, and which of the supported-magnitude and literal-survival requirements fails first. Every breakdown factor must be identified as observed-replication or projected. After the evidence source has been made explicit, unsuffixed notation may be used for readability. A possible statement is:

At the observed case mix, model A satisfied the order- K supported-magnitude and literal-survival requirements for replacement of model B . Under the symmetric capped empirical case-mix challenge, the estimated breakdown factor was $\hat{\Gamma}_{\text{ECM},A:B}^{\dagger} = [\text{value}]$, computed with numerical tolerance [value] and global-optimization gap [value]. Through that factor, no observation could receive more than [value] times its uniform within-class weight, and every admissible weighting retained effective support of at least [fraction] of each observed outcome class. The first failure involved [supported magnitude, literal survival, or both]. This is a finite-evidence robustness result over represented cases, not a confidence statement about unseen cases or contexts.

With several empirical evaluations, the statement should open, “Across M separately conducted empirical evaluations satisfying the declared regime.” With one resampled evaluation, it should open, “Under the declared fixed-model empirical-resampling procedure,” and the breakdown factor in the statement must be qualified as projected; Monte Carlo error and stability across seeds and values of J must also be reported.

F.9 Scope of the empirical challenge

The empirical target consists of redistributions of the case types contained in the evaluation. Every admissible mix recombines observed cases. A type absent from the evaluation receives weight zero at every Γ ; reaching it requires new data or an explicit extrapolation model. Individual reweighting

may also concentrate on sample-specific errors instead of a named scientific subpopulation. The breakdown factor therefore has no interpretation as the magnitude of a known deployment shift.

The AP construction has the same boundary. Prior standardization changes the class prior while retaining the observed class-conditional score behavior. Computational replication resamples the observed units. Empirical case-mix reweighting redistributes those units within outcome class. These procedures cannot reveal score behavior or failure mechanisms absent from the observed evaluations. Separate empirical evaluations can expose such variation when it falls within the declared regime.

No value encountered in the Γ search is interpreted as an external target population. The procedure starts from uniform empirical weights, increases a normalized concentration allowance according to a fixed symmetric rule, and reports where the declared replacement policy breaks. The empirical case-mix breakdown factor is a reproducible property of the paired comparison under the declared assessment. It measures robustness over represented cases and remains silent about absent case types and mechanisms.

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